

UNIT 1: READING, WRITING, COMPARING AND CALCULATING WHOLE NUMBERS UP TO 1,000,000

Key Unit Competence

By the end of this unit, you should be able to read, write, compare and perform calculations involving whole numbers up to **1,000,000**.

1.1 Place Value of Digits

What is place value?

Place value is the value of a digit according to its position in a number.

For example, consider:

654,321

Digit	Place	Value
6	Hundred thousands	600,000
5	Ten thousands	50,000
4	Thousands	4,000
3	Hundreds	300
2	Tens	20
1	Ones	1

Therefore:

$$654,321 = 600,000 + 50,000 + 4,000 + 300 + 20 + 1$$

For **1,000,000**, the digit 1 is in the **millions** place.

Important place-value order

From right to left:

Ones → Tens → Hundreds → Thousands → Ten thousands → Hundred thousands → Millions

Example

Find the place value and value of 7 in **472,563**.

7 is in the **ten-thousands place**.

Its value is:

$$7 \times 10,000 = 70,000$$

Revision Exercise 1

1. State the place value of 8 in **583,214**.
 2. What is the value of 6 in **264,731**?
 3. State the place value of 9 in **912,405**.
 4. Expand **735,428** according to place value.
 5. What is the place value of 1 in **1,000,000**?
-

1.2 Reading Numbers up to 1,000,000

Large numbers become easier to read when they are separated into groups of three digits from the right.

Example:

654,321

Separate it as:

654 | 321

The first group represents thousands.

Therefore:

654,321 = Six hundred fifty-four thousand, three hundred twenty-one.

Another example

825,234

is read as:

Eight hundred twenty-five thousand, two hundred thirty-four.

Reading 1,000,000

1,000,000 = One million

Remember

When reading a large number:

1. Start grouping digits from the right.
2. Put the digits into groups of three.
3. Read the group on the left first.
4. Add its place name, such as **thousand** or **million**.
5. Read the remaining group.

Revision Exercise 2

Write these numbers in words:

1. 345,216
2. 702,415
3. 910,304
4. 600,125
5. 999,999
6. 1,000,000

1.3 Writing Numbers in Figures

Numbers written in words can also be expressed using digits.

Example 1

Write in figures:

Four hundred twenty-three thousand, five hundred twelve.

Break it down:

- Four hundred twenty-three thousand = 423,000
- Five hundred = 500
- Twelve = 12

Therefore:

$$423,000 + 500 + 12 = 423,512$$

Example 2

Write in figures:

Eight hundred twenty-five thousand, two hundred thirty-four.

Answer:

825,234

Using expanded form

A number can also be written as the sum of its place values.

Example:

153,221

$$= 100,000 + 50,000 + 3,000 + 200 + 20 + 1$$

Revision Exercise 3

Write in figures:

1. Two hundred thirty-five thousand, four hundred twelve.
2. Seven hundred thousand, five hundred twenty.
3. Nine hundred ninety-nine thousand, nine hundred ninety-nine.
4. Six hundred eight thousand, forty-two.

5. One million.

1.4 Forming Numbers Using Digits

Different numbers can be formed by changing the positions of digits.

Suppose we have:

1, 3, 5 and 2

Some possible numbers are:

- 1,352
- 1,523
- 2,315
- 3,521
- 5,321

Changing the position of a digit changes its **place value**.

For example:

In **5,321**, 5 represents **5,000**.

In **3,521**, 5 represents **500**.

Greatest and smallest numbers

To form the **greatest number**, arrange the digits from greatest to smallest.

Example:

Digits: **7, 2, 9, 4**

Greatest number:

9,742

To form the **smallest number**, arrange them from smallest to greatest:

2,479

Revision Exercise 4

Using the digits **8, 5, 2, 6**:

1. Form the greatest possible number.
 2. Form the smallest possible number.
 3. Form any three other numbers.
 4. In your greatest number, state the place value of 5.
-

1.5 Comparing Whole Numbers

We use three symbols when comparing numbers:

> means **greater than**

< means **less than**

= means **equal to**

Examples

$$450,000 > 320,000$$

$$125,400 < 125,900$$

$$600,000 = 600,000$$

How to compare large numbers

Start comparing digits from the **left**.

Example:

Compare:

$$573,421 \text{ and } 571,999$$

Hundred thousands:

$$5 = 5$$

Ten thousands:

$$7 = 7$$

Thousands:

$$3 > 1$$

Therefore:

$$573,421 > 571,999$$

Important

If two numbers do not have the same number of digits, the number with more digits is usually greater.

Example:

$$99,999 < 100,000$$

Revision Exercise 5

Insert $>$, $<$ or $=$.

1. 345,620 ___ 345,602
2. 782,100 ___ 782,100
3. 199,999 ___ 200,000
4. 901,245 ___ 899,999
5. 567,340 ___ 576,340

1.6 Addition of Whole Numbers

Addition means combining two or more quantities.

Addition without carrying

Example:

$$272,142 + 203,512$$

Arrange according to place value:

$$\begin{array}{r} 272,142 \\ + 203,512 \\ \hline 475,654 \end{array}$$

Therefore:

$$272,142 + 203,512 = 475,654$$

Steps

1. Arrange numbers vertically.
 2. Make sure the same place values are in the same columns.
 3. Start from the ones.
 4. Add each column from right to left.
-

1.7 Addition with Carrying

Sometimes the sum of the digits in one column is **10 or more**.

In this case, we carry to the next place value.

Example

Calculate:

$$278,596 + 345,827$$

$$\begin{array}{r} 278,596 \\ + 345,827 \\ \hline 624,423 \end{array}$$

Work from right to left and carry whenever the sum in a column reaches 10 or more.

Adding three or more numbers

The same method is used.

Example:

$$\begin{array}{r}
 125,350 \\
 213,425 \\
 + 310,125 \\
 \hline
 648,900
 \end{array}$$

Revision Exercise 6

Calculate:

1. $430,526 + 323,250$
 2. $252,143 + 235,322$
 3. $235,450 + 126,375$
 4. $123,456 + 234,567 + 345,678$
 5. $425,760 + 132,890 + 210,450$
-

1.8 Subtraction of Whole Numbers

Subtraction means taking one quantity away from another or finding the difference between quantities.

Subtraction without borrowing

Example:

$$\begin{array}{r}
 785,642 \\
 - 243,521 \\
 \hline
 542,121
 \end{array}$$

Steps

1. Arrange the numbers according to place value.
 2. Start with the ones.
 3. Subtract from right to left.
-

1.9 Subtraction with Borrowing

We borrow when the upper digit is smaller than the digit being subtracted.

Example

Calculate:

$$874,450 - 583,750$$

The answer is:

$$290,700$$

When a digit is too small:

1. Borrow 1 from the next column on the left.
2. The borrowed 1 represents 10 units of the current place.
3. Complete the subtraction.
4. Continue towards the left.

Another example

$$1,000,000 - 482,000 = 518,000$$

Revision Exercise 7

Calculate:

1. $852,757 - 193,583$
 2. $710,534 - 40,203$
 3. $900,000 - 345,678$
 4. $625,300 - 214,125$
 5. $1,000,000 - 675,250$
-

1.10 Quick Multiplication by 5

A useful method is:

Multiply by 10 and divide by 2.

Example

$$324 \times 5$$

$$324 \times 10 = 3,240$$

$$3,240 \div 2 = 1,620$$

Therefore:

$$\mathbf{324 \times 5 = 1,620}$$

Exercise

Calculate mentally:

1. 240×5
 2. 325×5
 3. 412×5
 4. 620×5
-

1.11 Quick Multiplication by 9

To multiply by 9:

Multiply by 10, then subtract the original number.

Example

$$\mathbf{245 \times 9}$$

$$245 \times 10 = 2,450$$

$$2,450 - 245 = 2,205$$

Therefore:

$$\mathbf{245 \times 9 = 2,205}$$

1.12 Quick Multiplication by 11

One method is:

Multiply by 10 and add the original number.

Example

$$324 \times 11$$

$$324 \times 10 = 3,240$$

$$3,240 + 324 = 3,564$$

Therefore:

$$324 \times 11 = 3,564$$

1.13 Quick Multiplication by 50

To multiply by 50:

Multiply by 100 and divide by 2.

Example

$$246 \times 50$$

$$246 \times 100 = 24,600$$

$$24,600 \div 2 = 12,300$$

Therefore:

$$246 \times 50 = 12,300$$

1.14 Quick Multiplication by 25

Since 25 is one quarter of 100:

Multiply by 100 and divide by 4.

Example

$$324 \times 25$$

$$324 \times 100 = 32,400$$

$$32,400 \div 4 = 8,100$$

Therefore:

$$324 \times 25 = 8,100$$

1.15 Quick Multiplication by 19

Since:

$$19 = 20 - 1$$

Multiply by 20 and subtract the original number.

Example

$$125 \times 19$$

$$125 \times 20 = 2,500$$

$$2,500 - 125 = 2,375$$

Therefore:

$$125 \times 19 = 2,375$$

1.16 Quick Multiplication by 49

Since:

$$49 = 50 - 1$$

Example

$$120 \times 49$$

$$120 \times 50 = 6,000$$

$$6,000 - 120 = 5,880$$

Therefore:

$$120 \times 49 = 5,880$$

1.17 Quick Multiplication by 90

Since:

$$90 = 9 \times 10$$

Example

$$215 \times 90$$

$$215 \times 9 = 1,935$$

$$1,935 \times 10 = 19,350$$

Therefore:

$$215 \times 90 = 19,350$$

1.18 Quick Multiplication by 99

Since:

$$99 = 100 - 1$$

Example

$$325 \times 99$$

$$325 \times 100 = 32,500$$

$$32,500 - 325 = 32,175$$

Therefore:

$$325 \times 99 = 32,175$$

Revision Exercise 8

Calculate:

1. 315×9
 2. 214×11
 3. 420×25
 4. 312×50
 5. 225×19
 6. 144×49
 7. 315×90
 8. 245×99
-

1.19 Multiplication by a Three-Digit Number

When multiplying by a three-digit number:

1. Multiply by the ones.
2. Multiply by the tens.
3. Multiply by the hundreds.
4. Arrange each partial product according to place value.
5. Add the partial products.

Example

Calculate:

$$365 \times 241$$

Break 241 into:

$$200 + 40 + 1$$

Then:

$$365 \times 1 = 365$$

$$365 \times 40 = 14,600$$

$$365 \times 200 = 73,000$$

Add:

$$\begin{array}{r} 365 \\ 14,600 \\ + 73,000 \\ \hline 87,965 \end{array}$$

Therefore:

$$365 \times 241 = 87,965$$

Revision Exercise 9

Calculate:

1. 833×410
2. 581×611
3. 648×212
4. 349×247
5. 317×149

1.20 Division by a Two-Digit Number

Division means sharing a quantity into equal groups or determining how many times one number is contained in another.

The number being divided is called the **dividend**.

The number we divide by is called the **divisor**.

The answer is called the **quotient**.

Example

$$864 \div 24 = 36$$

This means:

$$24 \times 36 = 864$$

We can use multiplication to check a division answer.

1.21 Division with a Remainder

Sometimes a number cannot be divided exactly.

Example:

$$100 \div 9$$

$$9 \times 11 = 99$$

$$100 - 99 = 1$$

Therefore:

$$100 \div 9 = 11 \text{ remainder } 1$$

The remainder must always be **smaller than the divisor**.

Checking

$$\text{Divisor} \times \text{Quotient} + \text{Remainder} = \text{Dividend}$$

For the example:

$$9 \times 11 + 1 = 100$$

Revision Exercise 10

Calculate:

1. $936 \div 24$
2. $875 \div 25$
3. $728 \div 14$
4. $945 \div 22$
5. $783 \div 15$

State the remainder where necessary.

1.22 Solving Real-Life Problems

The four basic operations are:

- **Addition (+)** — combining quantities
- **Subtraction (−)** — finding a difference or taking away
- **Multiplication (×)** — repeated addition or finding totals of equal groups
- **Division (÷)** — sharing or grouping equally

Steps for solving word problems

Step 1: Read the problem carefully.

Step 2: Identify what is given.

Step 3: Identify what you are asked to find.

Step 4: Decide which operation to use.

Step 5: Perform the calculation.

Step 6: Check your calculation.

Step 7: Write the answer with the correct unit.

Example 1 — Addition

A shop sold **125,450** books in January and **214,350** books in February.

How many books were sold altogether?

Solution

$$125,450 + 214,350 = 339,800$$

Answer: 339,800 books

Example 2 — Subtraction

A store had **500,000 Frw** and spent **175,500 Frw**.

Money remaining:

$$500,000 - 175,500 = 324,500 \text{ Frw}$$

Example 3 — Multiplication

One bottle of juice costs **350 Frw**.

Find the cost of **112 bottles**.

$$350 \times 112 = 39,200 \text{ Frw}$$

Example 4 — Division

A school has **960 exercise books** to distribute equally among **32 pupils**.

$$960 \div 32 = 30$$

Each pupil receives **30 exercise books**.

UNIT 1 REVISION SUMMARY

Remember these important points:

- A digit's position determines its place value.
- **1,000,000** is read as **one million**.
- Large numbers can be separated into groups of three digits to make them easier to read.
- Use $>$, $<$ and $=$ to compare numbers.
- Always align place values when adding and subtracting.
- Carry when addition produces 10 or more in a place-value column.
- Borrow when the upper digit is too small during subtraction.

- Multiplication can be made faster by using known relationships such as $\times 9 = \times 10 -$ the original number.
- In long multiplication, multiply according to ones, tens and hundreds and then add the partial products.
- Division may be exact or may produce a remainder.
- A remainder must be smaller than the divisor.
- Always identify the correct operation before solving a word problem.

END-OF-UNIT REVISION EXERCISE

A. Place Value and Number Writing

1. What is the value of 7 in **573,421**?
2. State the place value of 8 in **684,250**.
3. Write **725,304** in words.
4. Write **Nine hundred forty-two thousand, three hundred sixteen** in figures.
5. Expand **805,472** according to place value.

B. Comparing Numbers

Insert $>$, $<$ or $=$.

6. 425,600 ___ 425,060
7. 999,999 ___ 1,000,000
8. 735,210 ___ 735,210
9. 601,245 ___ 610,245

C. Operations

10. $325,421 + 214,358$
11. $456,728 + 325,694$
12. $900,000 - 456,725$
13. $723,450 - 214,325$
14. 245×9
15. 312×25
16. 425×99
17. 324×215
18. $864 \div 24$
19. $965 \div 32$

D. Word Problems

20. A farmer harvested **125,450 kg** of maize in one season and **214,750 kg** in another season. How many kilograms did the farmer harvest altogether?
21. A school had **850,000 Frw**. It spent **375,500 Frw** on learning materials. How much money remained?
22. One exercise book costs **450 Frw**. How much will **125 exercise books** cost?
23. A school distributes **1,200 pencils** equally among **40 pupils**. How many pencils does each pupil receive?
24. A factory produced **425,600 bottles**. It sold **275,850 bottles**. How many bottles remained?
25. Explain one situation in your daily life where you may use:
 - addition,
 - subtraction,
 - multiplication,
 - division.

UNIT 2: ADDITION AND SUBTRACTION OF INTEGERS

Key Unit Competence

By the end of this unit, you should be able to solve problems involving the **addition and subtraction of positive and negative integers**.

2.0 Introduction

An **integer** is a whole number that can be:

- Positive: **+1, +2, +3, +4, ...**
- Negative: **-1, -2, -3, -4, ...**
- Zero: **0**

Integers are useful in real-life situations such as:

- temperature changes,
- gains and losses,
- debts and payments,
- positions above or below sea level,
- scores in games.

A number line helps us understand positive and negative numbers:

$-5 -4 -3 -2 -1 0 +1 +2 +3 +4 +5$

Numbers to the **right** are greater.

Numbers to the **left** are smaller.

2.1 Addition of Integers Using a Number Line

When adding integers on a number line:

- Start at the first integer.
- If you add a **positive integer**, move to the **right**.
- If you add a **negative integer**, move to the **left**.

Example 1

Calculate:

$$(-7) + (+10)$$

Start at -7 .

Move **10 steps to the right**.

You stop at **+3**.

Therefore:

$$(-7) + (+10) = +3$$

Example 2

Calculate:

$$(+4) + (-3)$$

Start at +4.

Move 3 steps to the left.

You stop at +1.

Therefore:

$$(+4) + (-3) = +1$$

Example 3

Calculate:

$$(-2) + (-3)$$

Start at -2.

Move 3 steps to the left.

You stop at -5.

Therefore:

$$(-2) + (-3) = -5$$

Revision Exercise 1

Use a number line to calculate:

1. $(-1) + (+3)$
 2. $(-9) + (+4)$
 3. $(-10) + (+5)$
 4. $(-2) + (-3)$
 5. $(+3) + (-4)$
 6. $(+10) + (-7)$
 7. $(-15) + (-12)$
-

2.2 Addition of Integers by Calculation

You do not always need a number line.

There are two important rules.

Rule 1: Same Signs

If the integers have the **same sign**:

1. Add their values.
2. Keep the common sign.

Examples

$$(-3) + (-4)$$

$$3 + 4 = 7$$

Both numbers are negative.

Therefore:

$$(-3) + (-4) = -7$$

Another example:

$$(+3) + (+4) = +7$$

Rule 2: Different Signs

If the integers have **different signs**:

1. Subtract the smaller value from the larger value.
2. Take the sign of the number with the greater value.

Example

$$(-3) + (+4)$$

$$4 - 3 = 1$$

The greater value, 4, is positive.

Therefore:

$$(-3) + (+4) = +1$$

Another example

$$(+3) + (-4)$$

$$4 - 3 = 1$$

The number with the greater value is -4 .

Therefore:

$$(+3) + (-4) = -1$$

Quick Summary

Operation	Rule
$(+) + (+)$	Add and keep +
$(-) + (-)$	Add and keep -
$(+) + (-)$	Subtract and use sign of greater value
$(-) + (+)$	Subtract and use sign of greater value

Revision Exercise 2

Calculate:

1. $(-8) + (+5)$
 2. $(-6) + (+2)$
 3. $(-20) + (+16)$
 4. $(-2) + (+10)$
 5. $(-3) + (-3)$
 6. $(-12) + (+6)$
 7. $(-11) + (+10)$
 8. $(+4) + (+4)$
 9. $(-12) + (+1)$
-

2.3 Subtraction of Integers Using a Number Line

When subtracting integers using a number line:

- Start at the first number.
- Subtracting a **positive integer** means moving to the **left**.
- Subtracting a **negative integer** means moving to the **right**.

Example 1

Calculate:

$$(-1) - (+5)$$

Start at -1 .

Move 5 steps to the left.

You stop at -6 .

Therefore:

$$(-1) - (+5) = -6$$

Example 2

Calculate:

$$(+8) - (+3)$$

Start at $+8$.

Move 3 steps to the left.

You stop at $+5$.

Therefore:

$$(+8) - (+3) = +5$$

Example 3

Calculate:

$$(-1) - (-3)$$

Subtracting a negative means moving to the right.

Start at -1 and move 3 steps right.

You stop at $+2$.

Therefore:

$$(-1) - (-3) = +2$$

Revision Exercise 3

Use a number line:

1. $(+8) - (+3)$
2. $(-6) - (+2)$
3. $(-8) - (+3)$
4. $(+7) - (+9)$
5. $(+7) - (+3)$
6. $(+9) - (+9)$

2.4 Subtraction of Integers by Calculation

The most important subtraction rule is:

Subtracting an integer means adding its opposite.

In symbols:

$$\mathbf{a - b = a + (-b)}$$

Therefore:

Subtracting a positive becomes adding a negative.

Subtracting a negative becomes adding a positive.

Example 1

$$(-6) - (+3)$$

Change subtraction to addition:

$$-6 + (-3)$$

$$= -9$$

Example 2

$$(+6) - (-3)$$

Change subtraction to addition:

$$+6 + (+3)$$

$$= +9$$

Example 3

$$4 - (-2)$$

$$= 4 + 2$$

$$= 6$$

Example 4

$$(+7) - (+3)$$

$$= +7 + (-3)$$

$$= +4$$

Easy Rule to Remember

Change subtraction to addition and change the sign of the second number.

Example:

$$8 - (-5)$$

becomes:

$$8 + 5 = 13$$

Revision Exercise 4

Calculate:

1. $(+12) - (+10)$
 2. $(-15) - (+5)$
 3. $(-7) - (-3)$
 4. $(-8) - (+2)$
 5. $(+16) - (-3)$
 6. $(-11) - (+6)$
 7. $(+7) - (+13)$
 8. $(-10) - (-8)$
 9. $(+9) - (-4)$
-

2.5 Additive Inverse

The **additive inverse** of a number is the number that gives **zero** when added to the original number.

It is also called the **opposite** of a number.

Examples

The additive inverse of **+5** is **-5** because:

$$(+5) + (-5) = 0$$

The additive inverse of **-9** is **+9** because:

$$(-9) + (+9) = 0$$

Therefore:

- Opposite of $+7 = -7$
- Opposite of $-12 = +12$
- Opposite of $+20 = -20$
- Opposite of $-30 = +30$

In general:

Additive inverse of $+a$ is $-a$.

Additive inverse of $-a$ is $+a$.

Revision Exercise 5

Write the additive inverse of:

1. -3
 2. -10
 3. -11
 4. -14
 5. -15
 6. $+4$
 7. $+25$
 8. -100
-

2.6 Solving Problems Involving Integers

Integer problems often involve words such as:

- increase,
- decrease,
- gain,
- loss,
- above,
- below,
- profit,
- debt,
- rise,
- fall.

Positive integers may represent:

- gains,
- increases,
- temperatures above zero,
- positions above a reference point.

Negative integers may represent:

- losses,
- decreases,
- debts,
- temperatures below zero,
- positions below a reference point.

Steps for Solving Integer Word Problems

1. Read the problem carefully.
2. Identify the positive and negative numbers.
3. Decide whether to add or subtract.
4. Write the mathematical expression.
5. Calculate.
6. Write the final answer.

Example 1: Position on a Number Line

A number is **8 steps from -1** and is greater than **$+6$** .

There are two possible positions:

-9 or $+7$

Since the number must be greater than **$+6$** :

Answer = $+7$

Example 2: Temperature Change

The morning temperature is **$+15^{\circ}\text{C}$** .

The evening temperature is **$+6^{\circ}\text{C}$** .

Difference:

$$15 - 6 = 9$$

Therefore, the temperature **decreased by 9°C** .

Example 3: Difference Between Positive and Negative Temperatures

Highest temperature:

+56°C

Lowest temperature:

-89°C

Difference:

$$56 - (-89)$$

$$= 56 + 89$$

$$= \mathbf{145^{\circ}\text{C}}$$

Therefore, the difference is **145°C**.

Example 4: Money

Suppose you owe **30 Frw**.

This can be represented as:

-30 Frw

Someone gives you **40 Frw**.

Your balance becomes:

$$\mathbf{-30 + 40 = +10}$$

After paying your debt, you remain with:

10 Frw

IMPORTANT INTEGER RULES

Addition

Same signs → Add and keep the sign

Example:

$$-5 + (-3) = -8$$

Different signs → Subtract and take the sign of the greater value

Example:

$$-5 + 8 = +3$$

Subtraction

Change subtraction into addition of the opposite:

$$a - b = a + (-b)$$

Example:

$$5 - (-4) = 5 + 4 = 9$$

Additive inverse

A number and its opposite always add to zero.

$$+8 + (-8) = 0$$

UNIT 2 REVISION SUMMARY

Remember:

- Integers include positive numbers, negative numbers and zero.
- Positive integers lie to the right of zero on a number line.
- Negative integers lie to the left of zero.
- Adding a positive number means moving right.

- Adding a negative number means moving left.
 - When adding numbers with the same sign, add and keep the sign.
 - When adding numbers with different signs, subtract and use the sign of the greater value.
 - Subtracting a positive number moves left.
 - Subtracting a negative number moves right.
 - To subtract an integer, add its opposite.
 - The additive inverse of a number is its opposite.
 - A number plus its additive inverse equals zero.
 - Integers are useful in temperature, money, debts, gains, losses and positions.
-

END-OF-UNIT REVISION EXERCISE

A. Addition of Integers

1. $(-7) + (+10)$
2. $(+5) + (-8)$
3. $(-12) + (-6)$
4. $(+15) + (+7)$
5. $(-20) + (+13)$
6. $(+17) + (-9)$

B. Subtraction of Integers

7. $(+9) - (+4)$
8. $(+6) - (-4)$
9. $(-8) - (+5)$
10. $(+12) - (+15)$
11. $(-10) - (-8)$
12. $(-7) - (-3)$

C. Additive Inverses

Write the additive inverse of:

13. $+7$
14. -5
15. $+12$
16. -18
17. $+25$

D. Word Problems

18. The temperature is $+10^{\circ}\text{C}$ and drops by 7°C . Find the final temperature.
19. A mountain climber is **300 m above sea level**. He descends **450 m**. Find his new position relative to sea level.
20. A person has a debt of **5,000 Frw** and receives **8,000 Frw**. After paying the debt, how much money remains?
21. The morning temperature is -4°C . It rises by 9°C . What is the new temperature?
22. A submarine is at **-120 m** relative to sea level. It rises by **75 m**. What is its new position?
23. A player's score is **+15 points**. The player loses **22 points**. What is the new score?
24. A number is **8 steps from -3** and is greater than 4. Find the number.
25. A number is **15 steps from +10** and is less than -4. Find the number.

UNIT 3: PRIME FACTORIZATION AND DIVISIBILITY TESTS

Key Unit Competence

By the end of this unit, you should be able to:

- prime factorize numbers,
 - find the **Lowest Common Multiple (LCM)**,
 - find the **Greatest Common Factor (GCF)**,
 - use divisibility tests,
 - express repeated prime factors using indices or powers.
-

3.0 Introduction

Prime factorization and divisibility help us understand how numbers are formed and how they can be divided.

They are useful when:

- simplifying fractions,
- dividing items equally,
- finding common multiples,
- solving problems involving repeated events,
- doing quick mental calculations.

3.1 Prime Numbers and Prime Factorization

What is a prime number?

A **prime number** is a number that has exactly **two different factors**:

1 and itself.

Examples:

2, 3, 5, 7, 11, 13

For example:

Factors of 7 are:

1 and 7

Therefore, 7 is a prime number.

A number such as 6 is not prime because it has more than two factors:

1, 2, 3 and 6

What is prime factorization?

Prime factorization means writing a number as a product of prime numbers.

Example 1

Prime factorize **40**.

Divide repeatedly by prime numbers:

$$40 \div 2 = 20$$

$$20 \div 2 = 10$$

$$10 \div 2 = 5$$

$$5 \div 5 = 1$$

Therefore:

$$40 = 2 \times 2 \times 2 \times 5$$

Example 2

Prime factorize **30**.

$$30 \div 2 = 15$$

$$15 \div 3 = 5$$

$$5 \div 5 = 1$$

Therefore:

$$30 = 2 \times 3 \times 5$$

Revision Exercise 1

Prime factorize:

1. 40
2. 120
3. 170
4. 80
5. 200
6. 320
7. 540
8. 670

3.2 Prime Factorization Using Indices

Repeated factors can be written using **indices** or **powers**.

For example:

$$2 \times 2 = 2^2$$

$$2 \times 2 \times 2 = 2^3$$

$$3 \times 3 \times 3 \times 3 = 3^4$$

The small number written above is called the **index** or **power**.

Example 1

Prime factorize 16.

$$16 = 2 \times 2 \times 2 \times 2$$

Therefore:

$$16 = 2^4$$

Example 2

Prime factorize 68.

$$68 = 2 \times 2 \times 17$$

Therefore:

$$68 = 2^2 \times 17$$

Example 3

Prime factorize 60.

$$60 = 2 \times 2 \times 3 \times 5$$

Therefore:

$$60 = 2^2 \times 3 \times 5$$

Revision Exercise 2

Prime factorize and write the answers using indices:

1. 27
 2. 75
 3. 36
 4. 76
 5. 98
 6. 48
-

3.3 Lowest Common Multiple (LCM)

A **multiple** of a number is obtained by multiplying that number by whole numbers.

For example, multiples of 4 are:

4, 8, 12, 16, 20, 24, ...

Multiples of 6 are:

6, 12, 18, 24, 30, ...

The common multiples are:

12, 24, ...

The smallest common multiple is called the **Lowest Common Multiple (LCM)**.

Therefore:

LCM of 4 and 6 = 12

Method 1: Listing Multiples

Example

Find the LCM of 2, 4 and 8.

Multiples of 2:

2, 4, 6, 8, 10, 12, 14, 16, ...

Multiples of 4:

4, 8, 12, 16, 20, ...

Multiples of 8:

8, 16, 24, ...

The smallest common multiple is:

8

Therefore:

$$\text{LCM}(2, 4, 8) = 8$$

Method 2: Prime Factorization

Example

Find the LCM of 12 and 18.

Prime factorize:

$$12 = 2^2 \times 3$$

$$18 = 2 \times 3^2$$

Take every prime factor using the **highest power**.

For 2, highest power = 2^2

For 3, highest power = 3^2

Therefore:

$$\text{LCM} = 2^2 \times 3^2$$

$$= 4 \times 9$$

$$= 36$$

Important LCM Rule

When using prime factorization:

Take the highest power of each prime factor.

Example

Find the LCM of 4, 6 and 8.

$$4 = 2^2$$

$$6 = 2 \times 3$$

$$8 = 2^3$$

Take:

$$2^3 \times 3$$

$$= 8 \times 3$$

$$= 24$$

Therefore:

$$\text{LCM}(4, 6, 8) = 24$$

Revision Exercise 3

Find the LCM of:

1. 2, 5 and 10
2. 5, 6 and 9
3. 2, 6 and 8
4. 4, 5 and 12
5. 4, 6 and 9
6. 6, 15 and 10

3.4 Applications of LCM

LCM is useful when different events repeat at regular intervals.

Example

Three traffic lights flash every:

- 30 seconds,
- 45 seconds,
- 60 seconds.

To know when they will flash together again, find:

LCM of 30, 45 and 60

Prime factorization:

$$30 = 2 \times 3 \times 5$$

$$45 = 3^2 \times 5$$

$$60 = 2^2 \times 3 \times 5$$

Take the highest powers:

$$2^2 \times 3^2 \times 5$$

$$= 4 \times 9 \times 5$$

$$= \mathbf{180 \text{ seconds}}$$

$$180 \text{ seconds} = \mathbf{3 \text{ minutes}}$$

Therefore, the lights flash together again after **3 minutes**.

3.5 Greatest Common Factor (GCF)

The **Greatest Common Factor (GCF)** is the greatest factor that divides two or more numbers exactly.

For example:

Factors of 12:

1, 2, 3, 4, 6, 12

Factors of 18:

1, 2, 3, 6, 9, 18

Common factors:

1, 2, 3, 6

The greatest is:

6

Therefore:

$$\mathbf{GCF(12, 18) = 6}$$

Finding GCF Using Prime Factorization

Example

Find the GCF of 24 and 36.

Prime factorize:

$$\mathbf{24 = 2^3 \times 3}$$

$$\mathbf{36 = 2^2 \times 3^2}$$

Identify the common prime factors:

2 and 3

Take the **smallest powers**:

$$2^2 \text{ and } 3^1$$

Therefore:

$$\mathbf{GCF = 2^2 \times 3}$$

$$= 4 \times 3$$

$$= \mathbf{12}$$

Important GCF Rule

When using prime factorization:

Take only the common prime factors and use their smallest powers.

Example

Find the GCF of 28, 42 and 56.

Prime factorization:

$$28 = 2^2 \times 7$$

$$42 = 2 \times 3 \times 7$$

$$56 = 2^3 \times 7$$

Common prime factors are:

2 and 7

Take the smallest powers:

$$2 \times 7 = 14$$

Therefore:

$$\text{GCF}(28, 42, 56) = 14$$

Revision Exercise 4

Find the GCF of:

1. 14, 20 and 36
2. 24, 36 and 40
3. 72, 84 and 108
4. 84, 140 and 224
5. 18 and 24
6. 36 and 48

LCM AND GCF: IMPORTANT DIFFERENCE

LCM

Lowest Common Multiple

GCF

Greatest Common Factor

LCM	GCF
Uses multiples	Uses factors
With prime factors, take highest powers	With prime factors, take smallest common powers
Often used for repeating events	Often used for equal grouping

3.6 Divisibility Test for 2

A number is divisible by **2** if its last digit is:

0, 2, 4, 6 or 8

These are even digits.

Examples

49,140

Last digit = 0

Therefore:

49,140 is divisible by 2.

90,712

Last digit = 2

Therefore:

90,712 is divisible by 2.

90,721

Last digit = 1

Therefore:

90,721 is not divisible by 2.

Exercise

Identify the numbers divisible by 2:

1. 4,480
 2. 6,429
 3. 5,258
 4. 21,224
 5. 49,242
 6. 15,504
-

3.7 Divisibility Test for 3

A number is divisible by **3** if the **sum of its digits is divisible by 3**.

Example 1

Is **1,824** divisible by 3?

Add the digits:

$$1 + 8 + 2 + 4 = 15$$

$$15 \div 3 = 5$$

Therefore:

1,824 is divisible by 3.

Example 2

Is **23,416** divisible by 3?

$$2 + 3 + 4 + 1 + 6 = 16$$

16 is not divisible by 3.

Therefore:

23,416 is not divisible by 3.

Revision Exercise

Test whether these numbers are divisible by 3:

1. 1,836
 2. 5,613
 3. 9,786
 4. 6,123
 5. 56,004
-

3.8 Divisibility Test for 4

A number is divisible by **4** if the number formed by its **last two digits** is divisible by 4.

Example 1

456,312

Last two digits = 12

$$12 \div 4 = 3$$

Therefore:

456,312 is divisible by 4.

Example 2

106,526

Last two digits = 26

26 is not divisible by 4.

Therefore:

106,526 is not divisible by 4.

Revision Exercise

Test for divisibility by 4:

1. 3,148
2. 98,541
3. 83,710
4. 201,084
5. 390,712
6. 873,008

3.9 Divisibility Test for 5

A number is divisible by **5** when its last digit is:

0 or 5.

Examples

56,480

Ends in 0.

Therefore, it is divisible by 5.

225,445

Ends in 5.

Therefore, it is divisible by 5.

741,024

Ends in 4.

Therefore, it is not divisible by 5.

Revision Exercise

Which numbers are divisible by 5?

1. 736,920
2. 878,945
3. 572,315
4. 563,759

5. 410,458
 6. 369,000
-

3.10 Divisibility Test for 6

A number is divisible by **6** if it is divisible by **both 2 and 3**.

Therefore, check:

1. Is the number even?
2. Is the sum of its digits divisible by 3?

If both answers are yes, the number is divisible by 6.

Example

Is **2,700** divisible by 6?

First test for 2:

Last digit = 0.

So it is divisible by 2.

Now test for 3:

$$2 + 7 + 0 + 0 = 9$$

9 is divisible by 3.

Therefore:

2,700 is divisible by 6.

Another Example

458,716

Last digit 6 means it is divisible by 2.

Add the digits:

$$4 + 5 + 8 + 7 + 1 + 6 = 31$$

31 is not divisible by 3.

Therefore:

458,716 is not divisible by 6.

Revision Exercise

Test whether the following are divisible by 6:

1. 216
 2. 144
 3. 928
 4. 93,621
 5. 48,780
 6. 56,800
-

3.11 Divisibility Test for 8

A number is divisible by **8** if the number formed by its **last three digits** is divisible by 8.

A number ending in **000** is also divisible by 8.

Example 1

404,320

Last three digits = 320

$$320 \div 8 = 40$$

Therefore:

404,320 is divisible by 8.

Example 2

202,072

Last three digits = 072 = 72

$$72 \div 8 = 9$$

Therefore:

202,072 is divisible by 8.

Example 3

323,638

638 is not exactly divisible by 8.

Therefore:

323,638 is not divisible by 8.

Revision Exercise

Test whether these numbers are divisible by 8:

1. 679,168
 2. 217,800
 3. 436,756
 4. 842,056
 5. 300,400
 6. 374,912
-

3.12 Divisibility Test for 9

A number is divisible by **9** if the **sum of its digits is divisible by 9**.

Example 1

64,737

$$6 + 4 + 7 + 3 + 7 = 27$$

$$27 \div 9 = 3$$

Therefore:

64,737 is divisible by 9.

Example 2

607,131

$$6 + 0 + 7 + 1 + 3 + 1 = 18$$

18 is divisible by 9.

Therefore:

607,131 is divisible by 9.

Example 3

128,000

$$1 + 2 + 8 + 0 + 0 + 0 = 11$$

11 is not divisible by 9.

Therefore:

128,000 is not divisible by 9.

Revision Exercise

Test whether these are divisible by 9:

1. 713,610
 2. 819,234
 3. 999,045
 4. 515,230
 5. 304,133
 6. 530,280
-

3.13 Divisibility Test for 10

A number is divisible by **10** when it ends with **0**.

Examples

49,140 → divisible by 10

199,000 → divisible by 10

872,930 → divisible by 10

447,861 → not divisible by 10 because it ends with 1.

Revision Exercise

Identify the numbers divisible by 10:

1. 874,930
2. 582,140
3. 529,900
4. 81,420
5. 555,355
6. 725,660
7. 554,740
8. 457,834

DIVISIBILITY RULES SUMMARY

Divisor	Rule
2	Last digit is 0, 2, 4, 6 or 8
3	Sum of digits is divisible by 3
4	Last two digits form a number divisible by 4
5	Last digit is 0 or 5
6	Number is divisible by both 2 and 3

Divisor	Rule
8	Last three digits form a number divisible by 8
9	Sum of digits is divisible by 9
10	Last digit is 0

UNIT 3 REVISION SUMMARY

Remember:

- A prime number has exactly two factors: **1 and itself**.
 - Prime factorization means expressing a number as a product of prime factors.
 - Repeated factors can be written using indices.
 - LCM is the smallest common multiple.
 - When finding LCM using prime factors, use the **highest powers**.
 - GCF is the greatest common factor.
 - When finding GCF using prime factors, use the **smallest powers of common prime factors**.
 - Divisibility rules help us know whether numbers can be divided exactly without carrying out full division.
-

END-OF-UNIT REVISION EXERCISE

A. Prime Numbers and Factorization

1. Define a prime number.
2. List the first five prime numbers.
3. Prime factorize 240.
4. Express 180 as a product of prime factors using indices.
5. Prime factorize 72.
6. Write 48 in index form.

B. LCM

7. Find the LCM of 6, 9 and 12.

8. Find the LCM of 4, 8 and 10.
9. Find the LCM of 12 and 18.
10. Two bells ring every 20 minutes and 30 minutes. After how many minutes will they ring together again?

C. GCF

11. Find the GCF of 48, 40 and 72.
12. Find the GCF of 24 and 36.
13. Find the GCF of 28, 42 and 56.
14. A teacher has 24 pencils and 36 notebooks and wants to make equal prize packs without leftovers. What is the greatest possible number of packs?

D. Divisibility

15. State the divisibility rule for 2.
16. State the divisibility rule for 3.
17. State the divisibility rule for 4.
18. State the divisibility rule for 5.
19. State the divisibility rule for 6.
20. State the divisibility rule for 8.
21. State the divisibility rule for 9.
22. State the divisibility rule for 10.

E. Test the Numbers

23. Is **891,484** divisible by 2?
24. Is **541,656** divisible by 2?
25. Is **891,480** divisible by 6?
26. Is **497,185** divisible by 6?
27. Is **456,312** divisible by 4?
28. Is **64,737** divisible by 9?
29. Is **872,930** divisible by 10?
30. Is **679,168** divisible by 8?

UNIT 4: EQUIVALENT FRACTIONS AND OPERATIONS

Key Unit Competence

By the end of this unit, you should be able to:

- explain equivalent fractions,
 - find equivalent fractions,
 - compare fractions,
 - add and subtract fractions with different denominators,
 - use the LCM to find a common denominator,
 - add and subtract mixed numbers,
 - solve word problems involving fractions.
-

4.0 Introduction

Equivalent fractions are fractions that look different but represent the same value or proportion of a whole.

For example:

$$\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$$

Equivalent fractions can be obtained by multiplying or dividing both the numerator and denominator by the same non-zero number.

4.1 Exploring and Comparing Fractions

A fraction represents part of a whole.

For example:

$$\frac{3}{8}$$

- 3 is the numerator.

- 8 is the denominator.

The denominator shows how many equal parts the whole has been divided into.

The numerator shows how many of those parts are being considered.

Comparing Fractions Using Models

Fraction strips, circles and shaded regions can help us compare fractions.

If one shaded region covers more of the same-sized whole, that fraction is greater.

If it covers less, that fraction is smaller.

Example 1

Compare:

$\frac{1}{2}$ and $\frac{2}{6}$

Half of a whole is greater than two-sixths of the same whole.

Therefore:

$\frac{1}{2} > \frac{2}{6}$

Example 2

Compare:

$\frac{5}{8}$ and $\frac{5}{12}$

The fractions have the same numerator.

Since eighths are larger pieces than twelfths:

$\frac{5}{8} > \frac{5}{12}$

Revision Exercise 1

Compare using $>$, $<$ or $=$:

1. $\frac{1}{4}$ ____ $\frac{3}{8}$

2. $\frac{1}{2}$ ____ $\frac{2}{6}$

3. $\frac{5}{8}$ ____ $\frac{5}{12}$

4. $\frac{2}{4}$ ____ $\frac{1}{2}$

5. $\frac{3}{6}$ ____ $\frac{4}{6}$

4.2 Meaning of Equivalent Fractions

Equivalent fractions represent the same amount even though their numerators and denominators are different.

For example:

$$\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$$

If equal-sized pieces of paper are shaded to represent these fractions, the shaded areas are equal.

Example

Compare:

$\frac{1}{3}$ and $\frac{2}{6}$

They cover the same amount of the whole.

Therefore:

$$\frac{1}{3} = \frac{2}{6}$$

Another example:

$$\frac{3}{4} = \frac{6}{8} = \frac{12}{16}$$

4.3 Finding Equivalent Fractions

To find an equivalent fraction:

Multiply the numerator and denominator by the same whole number.

Example 1

Find three equivalent fractions of:

$$\frac{3}{7}$$

Multiply by 2:

$$\frac{3}{7} \times \frac{2}{2} = \frac{6}{14}$$

Multiply by 3:

$$\frac{3}{7} \times \frac{3}{3} = \frac{9}{21}$$

Multiply by 4:

$$\frac{3}{7} \times \frac{4}{4} = \frac{12}{28}$$

Therefore:

$$\frac{3}{7} = \frac{6}{14} = \frac{9}{21} = \frac{12}{28}$$

Example 2

Find two equivalent fractions of:

$$\frac{4}{11}$$

$$\frac{4}{11} \times \frac{2}{2} = \frac{8}{22}$$

$$\frac{4}{11} \times \frac{3}{3} = \frac{12}{33}$$

Therefore:

$$4/11 = 8/22 = 12/33$$

Finding an Equivalent Fraction with a Given Denominator

Suppose we want an equivalent fraction of:

$$\frac{1}{2}$$

With denominator 20.

Ask:

$$2 \times ? = 20$$

$$20 \div 2 = 10$$

Multiply both numerator and denominator by 10:

$$\frac{1}{2} \times \frac{10}{10} = \frac{10}{20}$$

Therefore:

$$\frac{1}{2} = \frac{10}{20}$$

Another Example

Convert:

$$\frac{3}{10}$$

To a fraction with denominator 20.

$$10 \times 2 = 20$$

Therefore:

$$\frac{3}{10} \times \frac{2}{2} = \frac{6}{20}$$

Revision Exercise 2

Find two equivalent fractions for each:

1. $\frac{7}{9}$

2. $\frac{8}{12}$

3. $\frac{3}{5}$

4. $\frac{6}{7}$

5. $\frac{1}{5}$

6. $\frac{5}{12}$

4.4 Shading Equivalent Fractions

Equivalent fractions can also be shown using shaded models.

For example:

$$\frac{3}{8} = \frac{6}{16}$$

If a whole divided into 8 parts has 3 shaded parts, the same area can be represented by 6 shaded parts when the whole is divided into 16 equal parts.

Likewise:

$$\frac{5}{9} = \frac{10}{18}$$

And:

$$\frac{1}{2} = \frac{2}{4}.$$

4.5 Addition of Fractions with Different Denominators Using Models

Fractions with different denominators cannot be directly added until they represent equal-sized parts.

Example:

$$1/10 + 3/5$$

Change $3/5$ into tenths:

$$3/5 = 6/10$$

Then:

$$1/10 + 6/10 = 7/10$$

Therefore:

$$1/10 + 3/5 = 7/10$$

4.6 Addition of Fractions Using Equivalent Fractions

Case 1: One Denominator is a Multiple of the Other

Example:

$$1/3 + 4/9$$

The denominators are 3 and 9.

Since 9 is a multiple of 3, use 9 as the common denominator.

Convert:

$$1/3 = 3/9$$

Now add:

$$3/9 + 4/9 = 7/9$$

Therefore:

$$1/3 + 4/9 = 7/9$$

Steps

To add fractions with different denominators:

1. Identify the denominators.
2. Find a common denominator.
3. Convert the fractions into equivalent fractions.
4. Add the numerators.
5. Keep the common denominator.

6. Simplify if possible.

4.6.2 Using the LCM as the Common Denominator

When denominators are different, use their Lowest Common Multiple (LCM).

Example 1

Add:

$$\frac{1}{3} + \frac{2}{5}$$

LCM of 3 and 5 = 15

Convert:

$$\frac{1}{3} = \frac{5}{15}$$

$$\frac{2}{5} = \frac{6}{15}$$

Add:

$$\frac{5}{15} + \frac{6}{15} = \frac{11}{15}$$

Therefore:

$$1/3 + 2/5 = 11/15$$

Example 2

Add:

$$1/3 + 1/15$$

LCM of 3 and 15 = 15

Convert:

$$1/3 = 5/15$$

Then:

$$5/15 + 1/15 = 6/15$$

Simplify:

$$6/15 = 2/5$$

Example 3

Add:

$$\frac{2}{5} + \frac{2}{6}$$

LCM of 5 and 6 = 30

Convert:

$$\frac{2}{5} = \frac{12}{30}$$

$$\frac{2}{6} = \frac{10}{30}$$

Add:

$$\frac{12}{30} + \frac{10}{30} = \frac{22}{30}$$

Simplify:

$$\frac{22}{30} = \frac{11}{15}$$

Revision Exercise 3

Add:

$$1. \frac{1}{2} + \frac{3}{5}$$

2. $\frac{6}{7} + \frac{2}{3}$

3. $\frac{1}{3} + \frac{6}{7}$

4. $\frac{1}{2} + \frac{1}{3}$

5. $\frac{1}{2} + \frac{1}{5}$

6. $\frac{3}{4} + \frac{2}{5}$

7. $\frac{1}{6} + \frac{2}{7}$

8. $\frac{3}{5} + \frac{2}{7}$

4.7 Addition of Mixed Numbers

A mixed number contains a whole number and a fraction.

Example:

$2\frac{1}{3}$

To add mixed numbers, you can:

- convert them to improper fractions, or
- add the whole numbers and fractions separately.

The textbook gives both methods.

Method 1: Convert to Improper Fractions

To convert:

$$2 \frac{1}{3}$$

Multiply the whole number by the denominator:

$$2 \times 3 = 6$$

Add the numerator:

$$6 + 1 = 7$$

Therefore:

$$2 \frac{1}{3} = \frac{7}{3}$$

Example

Add:

$$1 \frac{1}{6} + 2 \frac{3}{6}$$

Convert:

$$1 \frac{1}{6} = \frac{7}{6}$$

$$2 \frac{3}{6} = 15/6$$

Add:

$$7/6 + 15/6 = 22/6$$

Simplify:

$$22/6 = 11/3 = 3 \frac{2}{3}$$

Method 2: Add Whole Numbers and Fractions Separately

Example:

$$5 \frac{3}{4} + 2 \frac{2}{3}$$

Whole numbers:

$$5 + 2 = 7$$

Fractions:

$$\frac{3}{4} + \frac{2}{3}$$

LCM of 4 and 3 = 12

$$\frac{3}{4} = \frac{9}{12}$$

$$\frac{2}{3} = \frac{8}{12}$$

So:

$$\frac{9}{12} + \frac{8}{12} = \frac{17}{12} = 1 \frac{5}{12}$$

Now:

$$7 + 1 \frac{5}{12} = 8 \frac{5}{12}$$

Revision Exercise 4

Add:

1. $3 \frac{3}{8} + 1 \frac{1}{4}$

2. $4 \frac{1}{5} + 3 \frac{1}{2}$

3. $2 \frac{2}{9} + 1 \frac{3}{4}$

4. $1 \frac{3}{5} + 2 \frac{1}{2}$

5. $2 \frac{1}{6} + 5 \frac{1}{3}$

4.8 Subtraction of Fractions with the Same Denominator

When denominators are the same:

1. subtract the numerators,
2. keep the denominator.

Example

$$15/24 - 4/24$$

$$= 11/24$$

Another example:

$$4/6 - 1/6$$

$$= 3/6$$

$$= 1/2$$

4.9 Subtraction of Fractions with Different Denominators

To subtract fractions with different denominators:

1. Find a common denominator, usually using the LCM.

2. Convert both fractions into equivalent fractions.
 3. Subtract the numerators.
 4. Keep the common denominator.
 5. Simplify if possible.
-

Example 1

Calculate:

$$7/8 - 3/4$$

Convert:

$$3/4 = 6/8$$

Then:

$$7/8 - 6/8 = 1/8$$

Example 2

Calculate:

$$7/8 - 2/3$$

LCM of 8 and 3 = 24

Convert:

$$7/8 = 21/24$$

$$2/3 = 16/24$$

Subtract:

$$21/24 - 16/24 = 5/24$$

Example 3

Calculate:

$$2/3 - 1/5$$

LCM of 3 and 5 = 15

Convert:

$$2/3 = 10/15$$

$$1/5 = 3/15$$

Subtract:

$$10/15 - 3/15 = 7/15$$

Revision Exercise 5

Subtract:

1. $7/8 - 1/2$

2. $4/5 - 1/10$

3. $5/9 - 1/3$

4. $1/3 - 1/6$

5. $9/10 - 3/6$

6. $7/8 - 1/3$

7. $8/9 - 1/5$

8. $6/7 - 2/5$

4.10 Subtraction of Mixed Numbers

There are three methods for subtracting mixed numbers:

1. subtract whole numbers and fractions separately,
2. borrow when necessary,

3. convert the mixed numbers into improper fractions.

Method 1: Subtract Separately

Example:

$$5 \frac{3}{4} - 2 \frac{1}{4}$$

Whole numbers:

$$5 - 2 = 3$$

Fractions:

$$\frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

Therefore:

$$5 \frac{3}{4} - 2 \frac{1}{4} = 3 \frac{1}{2}$$

Method 2: Borrowing

Suppose the fraction in the first mixed number is smaller than the fraction being subtracted.

Example:

$$4 \frac{1}{4} - 2 \frac{3}{4}$$

Since:

$$\frac{1}{4} < \frac{3}{4}$$

Borrow 1 from 4.

One whole = $\frac{4}{4}$.

Therefore:

$$4 \frac{1}{4} = 3 \frac{5}{4}$$

Now subtract:

$$3 \frac{5}{4} - 2 \frac{3}{4}$$

Whole numbers:

$$3 - 2 = 1$$

Fractions:

$$5/4 - 3/4 = 2/4$$

Therefore:

$$1 \frac{2}{4} = 1 \frac{1}{2}$$

Method 3: Improper Fractions

Example:

$$2 \frac{4}{5} - 1 \frac{3}{5}$$

Convert:

$$2 \frac{4}{5} = 14/5$$

$$1 \frac{3}{5} = 8/5$$

Subtract:

$$14/5 - 8/5 = 6/5$$

$$= 1 \frac{1}{5}$$

4.11 Word Problems Involving Addition of Fractions

When quantities represented by fractions are combined, use addition.

Example

During a meeting:

$\frac{1}{3}$ of the people were women.

$\frac{3}{5}$ were children.

Find the fraction representing women and children.

Add:

$$\frac{1}{3} + \frac{3}{5}$$

LCM of 3 and 5 = 15

$$\frac{1}{3} = \frac{5}{15}$$

$$\frac{3}{5} = \frac{9}{15}$$

Therefore:

$$5/15 + 9/15 = 14/15$$

So, women and children represented:

14/15 of the people.

Revision Exercise 6

1. Carene ate $\frac{4}{9}$ of a cake and her mother ate $\frac{1}{3}$. What fraction did they eat altogether?
 2. A cook mixed $\frac{1}{4}$ litre of milk and $\frac{1}{8}$ litre of water. Find the total quantity of liquid.
-

4.12 Word Problems Involving Subtraction of Fractions

Words that often indicate subtraction include:

- remaining,
- left,
- difference,
- take away,
- reduced by,
- decreased by,

- fewer than.

Example

A mother had:

$\frac{5}{9}$ litre of milk

After the baby drank some, there remained:

$\frac{1}{3}$ litre

How much did the baby drink?

Calculate:

$$\frac{5}{9} - \frac{1}{3}$$

Convert:

$$\frac{1}{3} = \frac{3}{9}$$

Therefore:

$$\frac{5}{9} - \frac{3}{9} = \frac{2}{9}$$

The baby drank:

$\frac{2}{9}$ litre.

Another Example

Jane has a stick $\frac{7}{8}$ m long.

Michael has a stick $\frac{5}{6}$ m long.

Difference:

LCM of 8 and 6 = 24

$$\frac{7}{8} = \frac{21}{24}$$

$$\frac{5}{6} = \frac{20}{24}$$

Therefore:

$$\frac{21}{24} - \frac{20}{24} = \frac{1}{24}$$

Jane's stick is $\frac{1}{24}$ m longer.

IMPORTANT FRACTION RULES

Equivalent Fractions

Multiply or divide the numerator and denominator by the same number.

Example:

$$\frac{2}{3} = \frac{4}{6}$$

Addition with Same Denominators

Add numerators:

$$\frac{2}{7} + \frac{3}{7} = \frac{5}{7}$$

Addition with Different Denominators

Find a common denominator first.

Example:

$$\frac{1}{2} + \frac{1}{3}$$

$$\text{LCM} = 6$$

$$\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$$

Subtraction with Same Denominators

Subtract numerators:

$$\frac{5}{8} - \frac{2}{8} = \frac{3}{8}$$

Subtraction with Different Denominators

Find a common denominator first.

Example:

$$\frac{3}{4} - \frac{1}{2}$$

$$\frac{3}{4} - \frac{2}{4} = \frac{1}{4}$$

UNIT 4 REVISION SUMMARY

Remember:

- The numerator is the top number of a fraction.
 - The denominator is the bottom number.
 - Equivalent fractions represent the same value.
 - Multiply the numerator and denominator by the same number to obtain equivalent fractions.
 - Fractions must have a common denominator before addition or subtraction.
 - The LCM of denominators can be used as the common denominator.
 - When denominators are the same, operate only on the numerators.
 - Simplify the final answer when possible.
 - Mixed numbers can be changed into improper fractions.
 - Borrowing can be used when subtracting mixed numbers.
 - Fractions are useful in sharing, measuring and many real-life situations.
-

END-OF-UNIT REVISION EXERCISE

A. Equivalent Fractions

1. Which fraction is equivalent to $\frac{1}{2}$?

- a) $\frac{2}{3}$
- b) $\frac{2}{4}$
- c) $\frac{3}{4}$

2. Write an equivalent fraction of $\frac{1}{3}$ with denominator 9.

3. Find an equivalent fraction of $\frac{4}{5}$ with denominator 15.

4. Write three equivalent fractions of $\frac{2}{7}$.

5. Fill in:

$$\frac{3}{4} = \frac{\quad}{12}$$

B. Addition

6. $\frac{1}{4} + \frac{1}{2}$

7. $\frac{3}{8} + \frac{1}{4}$

8. $\frac{1}{3} + \frac{2}{5}$

9. $\frac{3}{4} + \frac{2}{3}$

10. $\frac{5}{6} + \frac{1}{4}$

11. $2\frac{1}{3} + 1\frac{1}{2}$

12. $3\frac{3}{4} + 2\frac{1}{3}$

C. Subtraction

13. $5/8 - 3/8$

14. $7/8 - 1/2$

15. $3/4 - 1/3$

16. $5/6 - 1/4$

17. $2\frac{4}{5} - 1\frac{3}{5}$

18. $4\frac{1}{4} - 2\frac{3}{4}$

D. Word Problems

19. A cake was divided into 8 equal slices. Ruth ate 3 slices. What fraction of the cake did she eat?

20. Lily cut a cake into 12 equal slices. She ate 3 slices and her sister ate 4 slices. What fraction was eaten altogether?

21. A glass is filled to $\frac{3}{4}$. If $\frac{1}{4}$ is poured out, what fraction remains?

22. Martha walks $\frac{5}{8}$ km to school and Uwineza walks $\frac{1}{4}$ km. Who walks farther, and by how much?

23. A teacher has $\frac{3}{4}$ metre of thread and uses $\frac{1}{2}$ metre. How much thread remains?

24. A farmer has $\frac{7}{8}$ hectare of land. He plants crops on $\frac{1}{2}$ hectare. How much land remains?

UNIT 5: MULTIPLICATION AND DIVISION OF DECIMAL NUMBERS

Key Unit Competence

By the end of this unit, you should be able to:

- understand decimal numbers up to **three decimal places**,
- identify decimal place values,
- read and write decimals,
- compare decimal numbers,
- convert fractions to decimals and decimals to fractions,
- multiply and divide decimal numbers,
- solve real-life problems involving decimals.

5.0 Introduction

A **decimal number** has a whole-number part and a fractional part separated by a **decimal point**.

Examples:

2.5

0.75

14.236

Decimals are used in everyday life when dealing with measurements, temperatures, money, quantities and other calculations that require accuracy.

This unit deals mainly with:

- tenths,
 - hundredths,
 - thousandths,
 - decimal place value,
 - conversion between fractions and decimals,
 - multiplication,
 - division.
-

5.1 Decimal Fractions

A **decimal fraction** is a fraction whose denominator is a power of 10, such as:

10, 100, 1,000

Examples:

$1/10$ = one tenth

$1/100$ = one hundredth

$1/1,000$ = one thousandth

These fractions can be written as decimal numbers:

$1/10 = 0.1$

$1/100 = 0.01$

$1/1,000 = 0.001$

Tenths

When a whole is divided into **10 equal parts**, each part is called **one tenth**.

$1/10 = 0.1$

Examples:

$$2/10 = 0.2$$

$$4/10 = 0.4$$

$$7/10 = 0.7$$

Hundredths

When a whole is divided into **100 equal parts**, each part is called **one hundredth**.

$$1/100 = 0.01$$

Examples:

$$4/100 = 0.04$$

$$25/100 = 0.25$$

$$85/100 = 0.85$$

Thousandths

When a whole is divided into **1,000 equal parts**, each part is called **one thousandth**.

$$1/1,000 = 0.001$$

Examples:

$$5/1,000 = 0.005$$

$$256/1,000 = 0.256$$

$$705/1,000 = 0.705$$

5.1.1 Reading and Writing Decimal Numbers

Decimals can be read using the decimal point.

Examples:

0.02

is read as:

Zero point zero two

It also represents:

Two hundredths

0.3

is read:

Zero point three

It represents:

Three tenths

0.005

is read:

Zero point zero zero five

It represents:

Five thousandths

0.85

is read:

Zero point eight five

It represents:

Eighty-five hundredths.

Writing Decimals in Figures

Three hundredths:

0.03

Seven tenths:

0.7

Nine thousandths:

0.009

Four point nine:

4.9

Revision Exercise 1

Write in words:

1. 0.4
2. 0.08
3. 0.256
4. 0.007
5. 436.2
6. 196.261

Write in figures:

7. Five tenths
8. Six hundredths
9. Eight thousandths
10. Zero point two three five
11. Eight hundred seven thousandths

5.2 Place Value of Decimals

The digits after the decimal point have different place values.

The first digit after the decimal point is the **tenths**.

The second digit is the **hundredths**.

The third digit is the **thousandths**.

Place Value Table

Ones Decimal point Tenths Hundredths Thousandths

3 . 6 4 7

Therefore, in:

3.647

- 3 is in the **ones** place.
- 6 is in the **tenths** place.
- 4 is in the **hundredths** place.
- 7 is in the **thousandths** place.

Values of the Digits

In **3.647**:

6 tenths = **0.6**

4 hundredths = **0.04**

7 thousandths = **0.007**

Therefore:

$$\mathbf{3.647 = 3 + 0.6 + 0.04 + 0.007}$$

Example

Find the place value of 4 in:

356.4

The 4 is the first digit after the decimal point.

Therefore, 4 is in the **tenths place**.

Its value is:

0.4

Another Example

Find the place value of 4 in:

0.004

The 4 is the third digit after the decimal point.

Therefore, 4 is in the **thousandths place**.

Its value is:

0.004

Revision Exercise 2

State the place value of the indicated digit:

1. 4 in **356.4**
2. 4 in **236.254**
3. 4 in **0.004**
4. 4 in **0.245**
5. 2 in **56.235**
6. 2 in **43.325**
7. 2 in **0.002**

5.3 Comparing and Ordering Decimal Numbers

We use:

> greater than

< less than

= equal to

to compare decimals.

How to Compare Decimals

Step 1: Compare the whole-number parts

Example:

$$5.7 > 4.98$$

because:

$$5 > 4$$

Step 2: If the whole-number parts are equal, compare tenths

Example:

$$2.3 > 2.15$$

Write 2.3 as:

$$2.30$$

Compare:

$$2.30 \text{ and } 2.15$$

3 tenths $>$ 1 tenth.

Therefore:

$$2.3 > 2.15$$

Step 3: If tenths are equal, compare hundredths

Continue from left to right until the digits differ.

Adding Zeros

Zeros may be added to the right of a decimal without changing its value.

For example:

$$0.5 = 0.50 = 0.500$$

Similarly:

$$6 = 6.0 = 6.00$$

Examples

Compare:

$$0.2 \text{ and } 0.4$$

Therefore:

$$0.2 < 0.4$$

Compare:

$$0.05 \text{ and } 0.08$$

Therefore:

$$0.05 < 0.08$$

Compare:

$$0.009 \text{ and } 0.004$$

Therefore:

$$0.009 > 0.004$$

Order:

$$0.009, 0.04, 0.1$$

From smallest to largest:

$$0.009 < 0.04 < 0.1$$

Ordering Decimals

To arrange decimals:

1. Align the decimal points.
2. Add zeros where necessary.
3. Compare from left to right.
4. Arrange according to the required order.

Example

Arrange from smallest to largest:

0.5, 0.42, 0.08, 0.6

Write:

0.50

0.42

0.08

0.60

Therefore:

$0.08 < 0.42 < 0.5 < 0.6$

Revision Exercise 3

Insert $>$, $<$ or $=$.

1. 0.005 ___ 0.007

2. 3.40 ___ 3.040

3. 0.77 ___ 0.770

4. 0.825 ___ 0.826

5. 0.3 ___ 0.30

6. 0.09 ___ 0.1

Arrange from smallest to largest:

7. $0.01, 0.05, 0.02, 0.04$

8. $0.452, 0.252, 0.436$

9. $0.5, 0.4, 0.6, 0.8$

5.4 Conversion of Fractions to Decimals

To convert a fraction into a decimal:

Divide the numerator by the denominator.

Example 1

Convert:

$$\frac{3}{4}$$

$$3 \div 4 = \mathbf{0.75}$$

Therefore:

$$\frac{3}{4} = \mathbf{0.75}$$

Fractions with Denominators 10, 100 and 1,000

These can be converted directly using place value.

Examples:

$$\frac{7}{10} = \mathbf{0.7}$$

$$\frac{25}{100} = \mathbf{0.25}$$

$$\frac{8}{1,000} = \mathbf{0.008}$$

Method Using Equivalent Fractions

Example

Convert:

$$\frac{2}{5}$$

Make the denominator 10.

Multiply numerator and denominator by 2:

$$2/5 \times 2/2 = 4/10$$

Therefore:

$$2/5 = 0.4$$

Another Example

Convert:

$$15/25$$

Make the denominator 100.

Multiply by 4:

$$15/25 \times 4/4 = 60/100$$

Therefore:

$$15/25 = 0.60 = 0.6$$

Terminating and Repeating Decimals

A decimal that ends is called a **terminating decimal**.

Example:

$$1/2 = 0.5$$

A decimal that continues with a repeating pattern is called a **repeating decimal**.

Example:

$$1/3 = 0.3333...$$

Revision Exercise 4

Convert to decimals:

1. $5/10$

2. $\frac{6}{25}$
 3. $\frac{8}{1000}$
 4. $\frac{21}{25}$
 5. $\frac{2}{50}$
 6. $\frac{3}{8}$
 7. $\frac{1}{2}$
 8. $\frac{3}{4}$
-

5.5 Conversion of Decimals to Fractions

To convert a decimal into a fraction:

1. Identify the number of decimal places.
 2. Write the decimal over 10, 100 or 1,000.
 3. Simplify the fraction if possible.
-

One Decimal Place

Example:

0.5

There is one decimal place.

Therefore:

$$\mathbf{0.5 = \frac{5}{10}}$$

Simplify:

$$\mathbf{\frac{5}{10} = \frac{1}{2}}$$

Thus:

$$\mathbf{0.5 = \frac{1}{2}}$$

Two Decimal Places

Example:

0.75

There are two decimal places.

Therefore:

$$\mathbf{0.75 = 75/100}$$

Simplify:

$$\mathbf{75/100 = 3/4}$$

Therefore:

$$\mathbf{0.75 = 3/4}$$

Three Decimal Places

Example:

0.475

There are three decimal places.

Therefore:

$$\mathbf{0.475 = 475/1000}$$

Simplify:

$$\mathbf{475/1000 = 19/40}$$

Therefore:

$$\mathbf{0.475 = 19/40}$$

Another example:

$$\mathbf{0.89 = 89/100}$$

This fraction cannot be simplified further.

Revision Exercise 5

Convert to fractions and simplify:

1. 0.75
2. 0.455
3. 0.625
4. 0.075
5. 0.41
6. 0.5
7. 0.25
8. 0.125

5.6 Multiplication of Decimal Numbers

We can multiply:

- a decimal by a whole number,
- a decimal by another decimal.

To multiply decimal numbers:

1. Ignore the decimal points temporarily.
2. Multiply as whole numbers.
3. Count the total number of decimal places in the factors.
4. Put the same number of decimal places in the answer.

Example 1: Decimal $\times$ Whole Number

Calculate:

$$0.8 \times 4$$

Ignore the decimal:

$$8 \times 4 = 32$$

There is one decimal place in 0.8.

Therefore:

$$0.8 \times 4 = 3.2$$

Example 2: Decimal $\times$ Decimal

Calculate:

$$0.2 \times 0.7$$

Ignore decimals:

$$2 \times 7 = 14$$

There are:

- 1 decimal place in 0.2
- 1 decimal place in 0.7

Total = **2 decimal places**

Therefore:

$$0.2 \times 0.7 = 0.14$$

Example 3

Calculate:

$$0.4 \times 0.16$$

Ignore decimals:

$$4 \times 16 = 64$$

Total decimal places:

$$1 + 2 = 3$$

Therefore:

$$0.4 \times 0.16 = 0.064$$

Multiplication by 10, 100 and 1,000

Multiplying a decimal by powers of 10 moves the decimal point to the **right**.

$\times 10$

Move one place right.

$$0.45 \times 10 = 4.5$$

$\times 100$

Move two places right.

$$0.089 \times 100 = 8.9$$

$\times 1,000$

Move three places right.

$$0.006 \times 1,000 = 6$$

Revision Exercise 6

Calculate:

1. 0.06×7
2. 7.04×4
3. 2.66×11
4. 0.2×0.6
5. 0.14×0.2
6. 0.17×0.3
7. 0.089×100
8. 2.533×100
9. $0.006 \times 1,000$
10. $13.507 \times 1,000$

5.7 Division of Decimal Numbers

When dividing by a decimal, change the divisor into a whole number.

To do this:

- multiply by 10 if there is 1 decimal place,
- multiply by 100 if there are 2 decimal places,
- multiply by 1,000 if there are 3 decimal places.

Whatever is done to the divisor must also be done to the dividend.

Example 1

Calculate:

$$5 \div 0.2$$

Write:

$$5/0.2$$

Multiply both numbers by 10:

$$50/2$$

$$= 25$$

Therefore:

$$5 \div 0.2 = 25$$

Example 2

Calculate:

$$3.609 \div 0.03$$

There are two decimal places in 0.03.

Multiply both numbers by 100:

$$3.609 \times 100 = 360.9$$

$$0.03 \times 100 = 3$$

Now divide:

$$360.9 \div 3 = 120.3$$

Therefore:

$$3.609 \div 0.03 = 120.3$$

Another Example

Calculate:

$$6.4 \div 1.6$$

Multiply both numbers by 10:

$$64 \div 16 = 4$$

Therefore:

$$6.4 \div 1.6 = 4$$

Revision Exercise 7

Calculate:

1. $0.2 \div 5$
2. $4 \div 0.02$
3. $0.44 \div 1.1$
4. $1.792 \div 0.07$
5. $6.4 \div 1.6$
6. $2.4 \div 0.08$
7. $12.22 \div 26$
8. $0.13 \div 0.05$

5.8 Mixed Operations: Multiplication and Division

Some calculations contain both multiplication and division.

Example:

$$0.06 \times 0.2 \div 0.04$$

First multiply:

$$0.06 \times 0.2 = 0.012$$

Then divide:

$$0.012 \div 0.04$$

Multiply both by 100:

$$1.2 \div 4 = 0.3$$

Therefore:

$$0.06 \times 0.2 \div 0.04 = 0.3$$

Another Example

Calculate:

$$0.1 \times 0.36 \div 0.09$$

First:

$$0.36 \div 0.09 = 4$$

Then:

$$0.1 \times 4 = 0.4$$

Therefore:

$$0.1 \times 0.36 \div 0.09 = 0.4$$

Revision Exercise 8

Calculate:

1. $0.4 \times 0.2 \div 0.8$
 2. $5 \times 1.6 \div 0.08$
 3. $29.14 \times 9.2 \div 0.2$
 4. $0.04 \times 0.2 \div 0.4$
 5. $0.5 \times 0.2 \div 0.4$
-

5.9 Problems Involving Multiplication and Division of Decimals

Decimals are often used in real-life calculations involving:

- distance,
- mass,
- capacity,
- sharing,
- buying and selling,
- measurements.

When solving a problem:

1. Read the problem carefully.
 2. Identify the information given.
 3. Decide whether to multiply or divide.
 4. Calculate.
 5. Include the correct unit.
-

Example 1: Multiplication

A small bottle contains **0.3 litres** of milk.

How much milk is contained in **12 bottles**?

Multiply:

$$0.3 \times 12 = 3.6$$

Therefore:

12 bottles contain 3.6 litres of milk.

Example 2: Division

A roll of cloth is **540 m** long.

It is cut into pieces of **3.6 m** each.

How many pieces are obtained?

Calculate:

$$540 \div 3.6$$

Multiply both numbers by 10:

$$5,400 \div 36$$

$$= 150$$

Therefore:

150 pieces can be obtained.

Example 3

Each pupil receives **0.5 loaf** of bread.

There are **20 pupils**.

Total:

$$20 \times 0.5 = 10$$

Therefore:

10 loaves are needed.

Example 4

David drives **2.2 km per day** for 7 days.

Distance:

$$2.2 \times 7 = 15.4 \text{ km}$$

Therefore:

David travels 15.4 km.

Revision Exercise 9

1. 20 pupils are each given **0.5 loaf** of bread. How many loaves are needed?
2. One bottle contains **0.3 litres**. How many litres are contained in 3 bottles?
3. David travels **2.2 km per day**. How far does he travel in 7 days?
4. A person has **8.45 kg of coffee** and wants to put it into bags of **0.25 kg** each. How many complete bags can be filled?
5. A motorcycle covers **5.25 km** using 1 litre of fuel. How far will it travel using 1.5 litres?

IMPORTANT DECIMAL RULES

Decimal Place Values

0.1 = one tenth

0.01 = one hundredth

0.001 = one thousandth

Comparing Decimals

Align decimal points and compare digits from left to right.

Example:

$$0.65 = 0.650$$

Fraction to Decimal

Divide:

Numerator $\div$ **Denominator**

Example:

$$1/4 = 0.25$$

Decimal to Fraction

Use place value.

Example:

$$0.25 = 25/100 = 1/4$$

Multiplying Decimals

1. Multiply as whole numbers.
 2. Count the decimal places.
 3. Put the decimal point into the product.
-

Dividing Decimals

Make the **divisor a whole number** by moving its decimal point.

Move the decimal point in the dividend by the same number of places.

UNIT 5 REVISION SUMMARY

Remember:

- A decimal has a whole part and a fractional part separated by a decimal point.
 - The first three decimal places are **tenths, hundredths and thousandths**.
 - 0.1 means one tenth.
 - 0.01 means one hundredth.
 - 0.001 means one thousandth.
 - Zeros added to the right of a decimal do not change its value.
 - Fractions can be converted to decimals by division.
 - Decimals can be converted to fractions using place value.
 - When multiplying decimals, count the total decimal places in the factors.
 - When dividing by a decimal, first change the divisor into a whole number.
 - Decimals are important in measurements and everyday calculations.
-

END-OF-UNIT REVISION EXERCISE

A. Decimal Place Value

1. What is the place value of 5 in **264.235**?
 2. What is the place value of 2 in **0.253**?
 3. State the value of 6 in **12.067**.
 4. What is the place value of 8 in **0.008**?
-

B. Reading and Writing Decimals

5. Write **52.067** in words.
 6. Write in figures:
 - a) Six hundred sixty-seven thousandths
 - b) Seventy-two hundredths
 - c) One and one tenth
-

C. Comparing Decimals

Insert $>$, $<$ or $=$:

7. 0.081 ____ 0.095

8. 0.25 ____ 0.205

9. 0.65 ____ 0.650

10. 0.007 ____ 0.07

11. 3.4 ____ 3.40

D. Conversions

12. Write **0.236** as a fraction.

13. Convert **$\frac{1}{2}$** into a decimal.

14. Convert **$\frac{3}{4}$** into a decimal.

15. Convert **0.625** into a fraction and simplify.

16. Convert **0.25** into a fraction.

E. Multiplication and Division

17. Calculate:

0.23×0.23

18. Calculate:

$3 \div 0.3$

19. Calculate:

$1.44 \div 1.2$

20. Calculate:

$0.2 \times 0.5 \div 0.01$

21. Calculate:

2.4×0.6

22. Calculate:

$$4.5 \div 0.5$$

F. Ordering

23. Arrange from smallest to largest:

0.02, 0.85, 0.26

24. Arrange from greatest to smallest:

0.45, 0.405, 0.54, 0.5

G. Word Problems

25. Two oranges are shared equally among 20 people. What decimal part of an orange does each person receive?
26. One bottle contains **0.75 litres** of juice. How much juice is contained in 8 bottles?
27. A rope is **18.6 metres** long. It is cut into pieces of **0.6 m** each. How many pieces are obtained?
28. A pupil walks **1.25 km** to school each day. How far does the pupil walk in 5 days?
29. A container contains **6.4 litres** of water. The water is shared equally among 8 people. How much does each person receive?
30. Explain why accurate use of decimals is important when dealing with measurements or money.

UNIT 6: APPLICATION OF DIRECT PROPORTION

Key Unit Competence

By the end of this unit, you should be able to:

- explain direct proportion,
- solve problems involving direct proportion,
- use ratios,

- find the constant of proportionality,
 - use direct proportion in scale drawings,
 - apply direct proportion in practical situations.
-

6.0 Introduction

Direct proportion describes a relationship where two quantities increase or decrease together at the same rate.

For example:

If one exercise book costs 500 Frw:

- 1 book costs 500 Frw
- 2 books cost 1,000 Frw
- 3 books cost 1,500 Frw

As the number of books increases, the total cost also increases proportionally.

Direct proportion is useful in:

- shopping,
 - travelling,
 - sharing,
 - cooking,
 - scale drawings,
 - comparing quantities.
-

6.1 Meaning of Direct Proportion

Two quantities are **directly proportional** when:

- one increases and the other increases in the same way,
- one decreases and the other decreases in the same way.

Example 1

Suppose each pupil has **5 books**.

Pupils Books

1	5
2	10
3	15
4	20
5	25

If the number of pupils doubles, the number of books also doubles.

Therefore, the number of books is directly proportional to the number of pupils.

Example 2: Scale Drawing

On a drawing:

1 cm represents 10 km

Therefore:

2 cm represents 20 km

3 cm represents 30 km

4 cm represents 40 km

For 3 cm:

$$3 \times 10 \text{ km} = 30 \text{ km}$$

Therefore:

3 cm represents 30 km.

Example 3: Time

A person takes **30 minutes** to walk from home to school.

Going to school and returning home involves two journeys.

Therefore:

$$30 \times 2 = 60 \text{ minutes}$$

60 minutes = 1 hour.

Revision Exercise 1

1. Each pupil has 4 exercise books. How many exercise books do:
 - 2 pupils have?
 - 5 pupils have?
 - 10 pupils have?
 2. One water tank contains 1,500 litres. How many litres are contained in 6 tanks?
 3. There are 20 pupils, and each has 10 notebooks. Find the total number of notebooks.
 4. On a map, 1 cm represents 8 km. How many kilometres are represented by 5 cm?
-

6.2 Ratios and Direct Proportion

A **ratio** compares two quantities.

Ratios are written using a colon:

a : b

For example:

If one pupil has 2 books:

pupils : books = 1 : 2

A ratio in a direct proportion shows a constant relationship between quantities.

Example

Three children have 9 sweets.

Ratio of children to sweets:

$$3 : 9$$

Simplify by dividing both parts by 3:

$$3 \div 3 : 9 \div 3$$

$$= 1 : 3$$

This means each child gets 3 sweets.

For 4 children:

$$4 \times 3 = 12$$

Therefore:

4 children need 12 sweets.

Simplifying Ratios

To simplify a ratio:

1. Find a number that divides both quantities.
2. Divide both sides of the ratio by that number.

Example 1

Simplify:

$$8 : 24$$

Divide by 8:

$$1 : 3$$

Example 2

Simplify:

$$21 : 42$$

Divide by 21:

$$1 : 2$$

Example 3

Simplify:

$$9 : 27$$

Divide by 9:

$$1 : 3$$

Revision Exercise 2

Simplify:

1. $8 : 24$
2. $21 : 42$
3. $9 : 27$
4. $12 : 36$
5. $15 : 45$
6. $20 : 100$

Ratio Word Problems

Example

A mathematics book weighs **420 g** and a dictionary weighs **560 g**.

Ratio:

$$420 : 560$$

Divide both by 140:

3 : 4

Therefore, the ratio of the mathematics book to the dictionary is:

3 : 4

Revision Exercise 3

1. A farmer has 360 oranges and 120 mangoes. Find the ratio of mangoes to oranges.
 2. A game park has 120 giraffes and 360 antelopes. Find the ratio of giraffes to antelopes in simplest form.
 3. A school has 80 boys and 120 girls. Find the ratio of boys to girls in simplest form.
-

6.3 Constant of Proportionality

The **constant of proportionality** is the fixed number that connects two directly proportional quantities.

If:

$$n = k \times m$$

then **k** is the constant of proportionality.

We can also write:

$$k = n/m.$$

Example 1

If 6 books are given to 1 child:

$$\text{books : children} = 6 : 1$$

Constant:

$$k = 6 \div 1 = 6$$

For 7 children:

$$6 \times 7 = 42$$

Therefore:

7 children need 42 books.

Example 2

The ratio of red counters to blue counters is:

$$15 : 7$$

If there are 14 blue counters, find the number of red counters.

Since:

$$14 \div 7 = 2$$

Multiply 15 by 2:

$$15 \times 2 = 30$$

Therefore:

30 red counters are needed.

Example 3

There are 30 pupils.

Each pupil needs 4 sheets of paper.

One pupil $\rightarrow$ 4 sheets

30 pupils $\rightarrow 4 \times 30$

= 120 sheets

Therefore:

120 sheets of paper are needed.

Example 4

Three buses carry 75 passengers.

First find passengers per bus:

$$75 \div 3 = 25$$

Therefore:

1 bus carries 25 passengers.

For 4 buses:

$$25 \times 4 = 100$$

Therefore:

4 buses carry 100 passengers.

Revision Exercise 4

1. Five baskets contain 40 mangoes. How many mangoes are in one basket?
2. Seven boxes contain 84 books. How many books are in one box?
3. If one child needs 3 pencils, how many pencils are needed for 15 children?
4. Four buses carry 120 passengers. How many passengers will 7 buses carry if each bus carries the same number?

6.4 Scale Drawing and Direct Proportion

A **scale drawing** represents a real object using a smaller or larger drawing.

The relationship between the drawing and the real object remains proportional.

A **scale factor** connects the drawing size and the actual size.

One form used in the source is:

Scale factor = Real size $\div$ Drawing size

Example 1

Real size = 50 cm

Drawing size = 10 cm

Scale factor:

$$50 \div 10 = 5$$

Therefore, the scale factor is:

5

Example 2

Real size = 60 cm

Drawing size = 12 cm

Scale factor:

$$60 \div 12 = 5$$

Both drawings have the same scale factor, so they have been scaled proportionally.

Example 3

A drawing is 6 cm long.

The scale is **1 : 10**.

This means:

1 cm on drawing = 10 cm actual size.

Therefore:

$$6 \times 10 = 60 \text{ cm}$$

Actual size = 60 cm

Example 4

A model car is 15 cm long and the real car is 6 m long.

First use the same units:

$$6 \text{ m} = 600 \text{ cm}$$

Ratio:

$$15 : 600$$

Divide both by 15:

$$\mathbf{1 : 40}$$

Therefore, the model-to-real scale is:

$$\mathbf{1 : 40}$$

Revision Exercise 5

1. A drawing is 4 cm and represents an actual length of 40 cm. Find the scale factor.
 2. A drawing is 7 cm long with a scale of 1 : 10. Find the actual length.
 3. A model is 20 cm long and represents an object 400 cm long. Write the scale in simplest form.
 4. On a map, 3 cm represents 27 km. Find the distance represented by:
 - 6 cm
 - 10 cm
-

6.5 Problems Involving Ratios

When solving ratio problems:

1. Identify the quantities being compared.
2. Write the ratio.
3. Simplify if necessary.

4. If a total is given, add the parts of the ratio.
 5. Divide the total by the total number of parts.
 6. Multiply by each part to find the quantities.
 7. Make sure units are consistent.
-

Example 1

Red and blue beads are in the ratio:

3 : 5

There are 40 beads altogether.

Total parts:

$$3 + 5 = 8$$

Value of one part:

$$40 \div 8 = 5$$

Red beads:

$$3 \times 5 = 15$$

Blue beads:

$$5 \times 5 = 25$$

Therefore:

15 red beads and 25 blue beads.

Example 2

Flour and butter are in the ratio:

2 : 1

200 g of flour requires 100 g of butter.

If 400 g of flour is used:

$$400 \div 2 = 200$$

Therefore:

200 g of butter is needed.

Example 3

Goats and chickens are in the ratio:

3 : 5

There are 320 animals altogether.

Total parts:

$$3 + 5 = 8$$

One part:

$$320 \div 8 = 40$$

Goats:

$$3 \times 40 = 120$$

Chickens:

$$5 \times 40 = 200$$

Therefore:

120 goats and 200 chickens

Example 4

Maize and rice are in the ratio:

3 : 5

Their total mass is 96 kg.

Total parts:

$$3 + 5 = 8$$

One part:

$$96 \div 8 = 12 \text{ kg}$$

Maize:

$$3 \times 12 = 36 \text{ kg}$$

Rice:

$$5 \times 12 = 60 \text{ kg}$$

Therefore:

36 kg of maize and 60 kg of rice

Revision Exercise 6

1. Yellow and red beads are in the ratio 1 : 8. If there are 6 yellow beads, how many red beads are needed?
 2. The ratio of boys to girls is 2 : 3. There are 30 pupils altogether. Find:
 - number of boys,
 - number of girls.
 3. The ratio of cows to goats is 4 : 5. There are 81 animals. Find the number of cows and goats.
 4. Maize and beans are in the ratio 3 : 2. Their total mass is 75 kg. Find the mass of each.
-

6.6 Problems Involving Direct Proportion

In direct proportion:

both quantities increase or decrease together while their ratio remains constant.

A useful method is:

1. Find the value of **one unit**.

2. Multiply by the required number of units.
-

Example 1

Three speakers serve 7 districts.

How many speakers are needed for 42 districts?

42 is:

$$42 \div 7 = 6$$

times 7.

Therefore:

$$3 \times 6 = 18$$

18 speakers are needed.

Example 2

The ratio of adults to children is:

$$4 : 5$$

There are 400 adults.

Since:

$$4 \text{ parts} = 400$$

One part:

$$400 \div 4 = 100$$

Children:

$$5 \times 100 = 500$$

Therefore:

There are 500 children.

Example 3

Eight copies of a mathematics book weigh 480 g.

Find the weight of one copy.

$$480 \div 8 = 60$$

Therefore:

One copy weighs 60 g.

Example 4

A car travels 12 km using 1 litre of petrol.

How far will it travel using 3 litres?

$$12 \times 3 = 36$$

Therefore:

36 km

Example 5

Three minibuses carry 54 passengers.

One minibus carries:

$$54 \div 3 = 18 \text{ passengers}$$

Eight minibuses carry:

$$18 \times 8 = 144$$

Therefore:

144 passengers

Example 6

A truck uses 3 litres of petrol to travel 72 km.

How much petrol is needed for 648 km?

First find how many times 72 fits into 648:

$$648 \div 72 = 9$$

Therefore:

$$3 \times 9 = 27$$

27 litres of petrol are required.

IMPORTANT RULES

Direct Proportion

If one quantity doubles, the other also doubles.

If one quantity triples, the other also triples.

Ratio

A ratio compares quantities.

Example:

2 : 5

Simplifying a Ratio

Divide both parts by the same number.

Example:

12 : 18

Divide by 6:

2 : 3

Constant of Proportionality

If:

$$y = kx$$

then:

$$k = y \div x$$

Ratio with a Total

For a ratio:

a : b

1. Add: **a + b**
 2. Divide total by **a + b**
 3. Multiply by each ratio part.
-

UNIT 6 REVISION SUMMARY

Remember:

- Direct proportion means two quantities change together at a constant rate.

- Ratios compare two or more quantities.
 - Ratios can be simplified by dividing both parts by the same number.
 - The constant of proportionality is the fixed multiplier connecting proportional quantities.
 - Scale drawings use direct proportion between drawing size and actual size.
 - When a total and ratio are given, first calculate the total number of ratio parts.
 - Finding the value of one unit often makes direct-proportion problems easier.
 - Direct proportion is useful in shopping, transport, measurement, sharing and scale drawings.
-

END-OF-UNIT REVISION EXERCISE

A. Direct Proportion

1. Each pupil receives 5 exercise books. How many exercise books are needed for 30 pupils?
 2. One tank contains 1,200 litres. How much water is contained in 8 tanks?
 3. One car travels 15 km using 1 litre of petrol. How far will it travel using 6 litres?
 4. Five workers produce 100 identical items when each produces the same number. How many items would 8 workers produce at that same rate?
-

B. Ratios

5. Simplify:

20 : 100

6. Simplify:

18 : 54

7. Find the ratio of 20 mango trees to 100 coffee trees.
 8. Find the ratio of 28 passengers to 64 passengers in simplest form.
-

C. Ratio Problems

9. Red and blue balls are in the ratio 3 : 5. There are 64 balls altogether. Find the number of each colour.
 10. Maize and rice are in the ratio 3 : 5 and have a total mass of 96 kg. Find the mass of each.
 11. Boys and girls are in the ratio 4 : 5. There are 45 pupils. Find the number of boys and girls.
-

D. Constant of Proportionality

12. Three buses carry 75 passengers. Find:
 - passengers per bus,
 - passengers carried by 6 buses.
 13. Four pupils need 16 pieces of paper. How many pieces does one pupil need?
 14. If one child receives 6 books, how many books are needed for 12 children?
-

E. Scale Drawing

15. A drawing measures 8 cm while the real object measures 40 cm. Find the scale factor.
 16. A map uses a scale where 1 cm represents 10 km. Find the actual distance represented by 7 cm.
 17. A model measures 10 cm while the actual object measures 200 cm. Write the scale as a simplified ratio.
-

F. Word Problems

18. Two loaves of bread are served to 8 children. How many loaves are required for 64 children?
19. A pupil drinks 5 glasses of water every day. How many glasses will the pupil drink in 10 days?
20. It takes 40 minutes to walk to the market. Find the total time needed to walk to the market and back home every day for 7 days.
21. Fifteen boys have a total mass of 300 kg and have equal masses. Find the mass of one boy.
22. One aircraft carries 100 passengers. How many passengers are carried by 6 identical aircraft?

23. A cleaner uses 8 litres of water every day. How many days will 64 litres last?
24. A baby drinks 4 glasses of milk every day. How many days will 132 glasses last?

UNIT 7: SOLVING PROBLEMS INVOLVING MEASUREMENTS OF LENGTH, CAPACITY AND MASS

Key Unit Competence

By the end of this unit, you should be able to:

- convert units of **length, capacity and mass**,
 - solve problems involving measurements,
 - calculate the number of intervals and objects on open and closed lines,
 - apply measurement knowledge in real-life situations.
-

7.0 Introduction

Measurement helps us describe quantities such as:

- **Length** — how long, tall or far something is.
- **Capacity** — how much liquid a container can hold.
- **Mass** — how heavy something is.

Measurements are used in many daily situations, such as cooking, travelling, shopping, construction and sports.

7.1 Revision on Length, Capacity and Mass

A. Units of Length

The metric units of length are:

km → hm → dam → m → dm → cm → mm

Each neighbouring unit changes by a factor of **10**.

Important relationships include:

$$1 \text{ km} = 1,000 \text{ m}$$

$$1 \text{ hm} = 100 \text{ m}$$

$$1 \text{ dam} = 10 \text{ m}$$

$$1 \text{ m} = 10 \text{ dm}$$

$$1 \text{ m} = 100 \text{ cm}$$

$$1 \text{ m} = 1,000 \text{ mm}$$

$$1 \text{ dm} = 10 \text{ cm} = 100 \text{ mm}$$

Converting Larger Units to Smaller Units

Multiply.

Example 1

Convert:

3 km to metres

$$1 \text{ km} = 1,000 \text{ m}$$

$$3 \text{ km} = 3 \times 1,000$$

$$= \mathbf{3,000 \text{ m}}$$

Example 2

Convert:

4 m to centimetres

$$1 \text{ m} = 100 \text{ cm}$$

$$4 \times 100 = \mathbf{400 \text{ cm}}$$

Converting Smaller Units to Larger Units

Divide.

Example

Convert:

500 cm to metres

$$100 \text{ cm} = 1 \text{ m}$$

$$500 \div 100 = \mathbf{5 \text{ m}}$$

Length Conversion Table

Unit	Symbol
kilometre	km
hectometre	hm
decametre	dam
metre	m
decimetre	dm
centimetre	cm
millimetre	mm

B. Units of Capacity

Capacity measures liquids.

The units are:

$\text{k}\ell \rightarrow \text{h}\ell \rightarrow \text{da}\ell \rightarrow \ell \rightarrow \text{d}\ell \rightarrow \text{c}\ell \rightarrow \text{m}\ell$

Important relationships:

$$1 \ell = 10 \text{ d}\ell$$

$$1 \ell = 100 \text{ c}\ell$$

$$1 \ell = 1,000 \text{ m}\ell$$

$$1 \text{ da}\ell = 10 \ell$$

Examples

Convert:

3 ℓ to $\text{m}\ell$

$$3 \times 1,000 = 3,000 \text{ m}\ell$$

Convert:

2,500 $\text{m}\ell$ to litres

$$2,500 \div 1,000 = 2.5 \ell$$

C. Units of Mass

Mass measures how heavy an object is.

The metric units include:

$\text{kg} \rightarrow \text{hg} \rightarrow \text{dag} \rightarrow \text{g} \rightarrow \text{dg} \rightarrow \text{cg} \rightarrow \text{mg}$

Important relationships:

$$1 \text{ kg} = 1,000 \text{ g}$$

$$1 \text{ g} = 10 \text{ dg}$$

$$1 \text{ dag} = 10 \text{ g}$$

$$1 \text{ g} = 1,000 \text{ mg}$$

Also:

$$1 \text{ tonne} = 1,000 \text{ kg}$$

Examples

Convert:

5 kg to grams

$$5 \times 1,000 = \mathbf{5,000 \text{ g}}$$

Convert:

2,500 g to kilograms

$$2,500 \div 1,000 = \mathbf{2.5 \text{ kg}}$$

Important Measurement Rules

When solving measurement problems:

1. Convert quantities to the **same unit**.
 2. Use a conversion table when necessary.
 3. Identify the important values and units.
 4. Always write the correct unit in the final answer.
-

Worked Examples

Example 1

Convert **30 mm** to centimetres.

$$10 \text{ mm} = 1 \text{ cm}$$

$$30 \div 10 = \mathbf{3 \text{ cm}}$$

Example 2

Convert **60 mm** to centimetres.

$$60 \div 10 = \mathbf{6 \text{ cm}}$$

Example 3

Convert **0.7 km** to centimetres.

$$1 \text{ km} = 100,000 \text{ cm}$$

$$0.7 \times 100,000 = \mathbf{70,000 \text{ cm}}$$

Example 4: Mixed Units

Subtract:

2 m 6 dm 4 cm from 9 m

Give the answer in decimetres.

Convert:

$$2 \text{ m} = 20 \text{ dm}$$

$$6 \text{ dm} = 6 \text{ dm}$$

$$4 \text{ cm} = 0.4 \text{ dm}$$

Therefore:

$$2 \text{ m } 6 \text{ dm } 4 \text{ cm} = \mathbf{26.4 \text{ dm}}$$

$$9 \text{ m} = \mathbf{90 \text{ dm}}$$

Subtract:

$$90 - 26.4 = 63.6 \text{ dm}$$

Revision Exercise 1

Convert:

1. 260 cm to metres
 2. 4,000 cm to metres
 3. 6 km to metres
 4. 60 cm to metres
 5. 30 dℓ to litres
 6. 1,050 mℓ to litres
 7. 2,500 mℓ to litres
 8. 3,450 kg to tonnes
 9. 2,050 kg to tonnes
 10. 170,000 kg to tonnes
-

7.2 Number of Intervals Between Objects on an Open Line

An **interval** is the fixed distance between two neighbouring objects.

Objects may include:

- trees,
- electric poles,
- pupils,
- fence posts,
- markings,
- strides.

On an **open line**, the relationship between the number of objects and intervals depends on whether spaces are left at the ends.

Case 1: No Interval is Left at Either End

Suppose there are 5 trees:

Tree — Tree — Tree — Tree — Tree

There are **4 intervals**.

Therefore:

Number of intervals = Number of objects – 1

and:

Number of objects = Number of intervals + 1

If distance and interval are known:

Number of objects = Distance ÷ Interval + 1

Example

Trees are planted every **4 m** along a distance of **16 m**, with a tree at both ends.

Number of intervals:

$$16 \div 4 = 4$$

Number of trees:

$$4 + 1 = \mathbf{5 \text{ trees}}$$

Case 2: One Interval Space is Left at One End

In this case:

Number of intervals = Number of objects

Therefore:

Number of objects = Distance $\div$ Interval

Example

A distance is 15 m.

Objects are placed every 5 m, but one interval is left at one end.

Number of objects:

$$15 \div 5 = \mathbf{3 \text{ objects}}$$

Case 3: Interval Spaces are Left at Both Ends

In this case:

Number of intervals = Number of objects + 1

Therefore:

Number of objects = Number of intervals - 1

or:

Number of objects = Distance $\div$ Interval - 1

Summary of Open-Line Rules

Situation	Rule
Object at both ends	Objects = intervals + 1
Space at one end	Objects = intervals
Space at both ends	Objects = intervals - 1

Stride Problems

A stride can also be treated as an interval.

Example

A person's stride is **10 dam**.

Distance travelled = **10 dam**

Convert:

10 dam = **1,000 dm**

Number of strides:

$1,000 \div 10$

= **100 strides**

Another Example

A person's stride is **90 cm**.

Distance = **27 dam**

Convert:

27 dam = 270 m

270 m = 27,000 cm

Number of strides:

$27,000 \div 90$

= **300 strides**

Revision Exercise 2

1. A road is 2 km long. Trees are planted 2 m apart. One interval is left at one end. How many trees are planted?
 2. A straight path is 50 m long. Posts are placed every 5 m with a post at both ends. How many posts are needed?
 3. A 30 m line has poles every 5 m, with spaces left at both ends. How many poles are used?
 4. An athlete has a stride of 80 cm and runs 80 dam. Find the number of strides.
 5. There are 91 trees in a straight line, with 8 m between consecutive trees. Find the total distance from the first tree to the last.
-

7.3 Number of Intervals on a Closed Line

A **closed line** forms a complete shape such as:

- a circle,
- square,
- rectangle,
- closed fence.

On a closed line:

Number of intervals = Number of poles or objects.

This is because the last object connects back to the first object.

Closed-Line Formula

For a closed line:

Number of intervals = Number of poles

Also:

Distance around the closed line = Number of intervals $\times$ Interval length

Therefore:

Number of poles = Perimeter $\div$ Interval length

and:

Interval length = Perimeter $\div$ Number of poles

Example 1: Rectangular Land

A rectangular piece of land measures:

48 m by 12 m

Poles are placed every **2 m**.

Step 1: Find the perimeter

Perimeter:

$$2 \times (48 + 12)$$

$$= 2 \times 60$$

$$= \mathbf{120\ m}$$

Step 2: Find the number of poles

$$120 \div 2 = \mathbf{60}$$

Therefore:

60 poles are needed.

Example 2: Square Plot

A square has sides of **50 m**.

Poles are placed every **2 m**.

Perimeter:

$$4 \times 50 = \mathbf{200 \text{ m}}$$

Number of poles:

$$200 \div 2 = \mathbf{100 \text{ poles}}$$

Example 3: Circular Fishpond

$$\text{Circumference} = \mathbf{154 \text{ m}}$$

$$\text{Interval} = \mathbf{3.5 \text{ m}}$$

Number of poles:

$$154 \div 3.5 = \mathbf{44 \text{ poles}}$$

Example 4: Finding Interval Length

A rectangular tank has:

$$\text{Length} = 125 \text{ m}$$

$$\text{Width} = 100 \text{ m}$$

$$\text{Number of poles} = 150$$

First find perimeter:

$$2 \times (125 + 100)$$

$$= 2 \times 225$$

$$= \mathbf{450 \text{ m}}$$

Interval:

$$450 \div 150$$

$$= 3 \text{ m}$$

Therefore:

Each interval is 3 m long.

Example 5

The distance around an animal park is:

$$4.2 \text{ km}$$

There are **210 trees** placed at equal intervals.

Convert:

$$4.2 \text{ km} = 4,200 \text{ m}$$

Since it is a closed line:

$$\text{Number of intervals} = 210$$

Interval length:

$$4,200 \div 210$$

$$= 20 \text{ m}$$

Therefore:

The trees are 20 m apart.

IMPORTANT FORMULAS

Length

$$\text{Distance} = \text{Number of intervals} \times \text{Interval length}$$

Open Line — Objects at Both Ends

$$\text{Objects} = \text{Distance} \div \text{Interval} + 1$$

Open Line — Space at One End

$$\text{Objects} = \text{Distance} \div \text{Interval}$$

Open Line — Spaces at Both Ends

$$\text{Objects} = \text{Distance} \div \text{Interval} - 1$$

Closed Line

$$\text{Objects} = \text{Distance around} \div \text{Interval}$$

because:

$$\text{Objects} = \text{Intervals}$$

UNIT 7 REVISION SUMMARY

Remember:

- Length measures how long or far something is.
 - Capacity measures how much liquid a container can hold.
 - Mass measures how heavy something is.
 - Always convert quantities to the same unit before calculating.
 - An interval is the distance between neighbouring objects.
 - On an open line with objects at both ends, intervals are one fewer than objects.
 - With a space at one end, intervals equal objects.
 - With spaces at both ends, intervals are one more than objects.
 - On a closed line, the number of intervals equals the number of objects.
 - Perimeter is used when objects are arranged around a closed field or shape.
-

END-OF-UNIT REVISION EXERCISE

A. Conversions

1. Convert **25 dm** to hectometres.
 2. Convert **300 mm** to decametres.
 3. Convert **3.45 tonnes** to kilograms.
 4. Convert **30 dℓ** to litres.
 5. Convert **2.6 km** to metres.
 6. Convert **4,500 mL** to litres.
 7. Convert **7.2 kg** to grams.
-

B. Measurement Problems

8. Calculate:

$$4.5 \text{ kg} + 13.6 \text{ dag}$$

Give the answer in kilograms.

9. Calculate:

$$4 \text{ hℓ} - 20 \text{ daℓ}$$

Give the answer in litres.

10. Calculate:

$$2 \text{ dam } 3 \text{ m} \times 5$$

Give the answer in hectometres.

C. Open-Line Intervals

11. A road is 100 m long. Trees are planted every 10 m with trees at both ends. How many trees are planted?
12. A path is 60 m long. Posts are placed every 6 m, but one interval is left at one end. How many posts are used?
13. A line is 45 m long. Poles are placed every 5 m, with spaces at both ends. How many poles are used?

14. There are 21 trees planted in a straight line 4 m apart. Find the distance from the first tree to the last.
-

D. Closed-Line Intervals

15. A square plot has sides of 40 m. Poles are placed every 5 m around it. How many poles are needed?
16. A rectangular field measures 60 m by 40 m. Posts are placed every 4 m around the field. Find the number of posts.
17. A circular pond has a circumference of 180 m. Poles are placed every 6 m. How many poles are required?
18. A closed fence is 600 m around and has 120 poles equally spaced. Find the distance between neighbouring poles.
-

E. Word Problems

19. An athlete runs **80 dam**. Her stride is **80 cm**. How many strides does she take?
20. A straight row contains **91 trees**, with **8 m** between consecutive trees. Find the total distance from the first tree to the last.
21. A rectangular plot is 80 m long and 40 m wide. It is fenced with posts 3 m apart. Find the number of posts if they are placed around the complete closed boundary.
22. Explain why standard measurement units are important in construction, transport or sports.

UNIT 8: SOLVING PROBLEMS INVOLVING TIME INTERVALS

Key Unit Competence

By the end of this unit, you should be able to:

- identify units of time,
- convert hours, minutes, seconds and days,
- find time intervals,
- add and subtract time,

- solve real-life problems involving time,
 - use time correctly in daily activities.
-

8.0 Introduction

Time is used every day.

We use time to know:

- when to wake up,
- when to go to school,
- how long a lesson lasts,
- how long a journey takes,
- when an activity starts,
- when an activity ends.

A **time interval** is the amount of time between two events.

Understanding time helps us organise and manage our daily activities.

Important Units of Time

Remember these relationships:

1 minute = 60 seconds

1 hour = 60 minutes

1 hour = 3,600 seconds

1 day = 24 hours

8.1 Converting Hours into Minutes

Since:

1 hour = 60 minutes

To change hours into minutes:

Multiply the number of hours by 60.

Formula

Minutes = Hours $\times$ 60

Example 1

Convert:

3 hours into minutes

$3 \times 60 = \mathbf{180 \text{ minutes}}$

Example 2

A tourist spends 6 hours visiting an animal park.

Find the time in minutes.

$6 \times 60 = \mathbf{360 \text{ minutes}}$

Therefore:

6 hours = 360 minutes.

Example 3

Convert:

2½ hours into minutes

2 hours = 120 minutes

$\frac{1}{2}$ hour = 30 minutes

Therefore:

$2\frac{1}{2}$ hours = 150 minutes

Revision Exercise 1

Convert into minutes:

1. 2 hours
 2. 5 hours
 3. 8 hours
 4. 14 hours
 5. $\frac{1}{2}$ hour
 6. $1\frac{1}{2}$ hours
 7. $3\frac{1}{2}$ hours
 8. $6\frac{1}{4}$ hours
-

8.2 Converting Minutes into Hours

Since:

60 minutes = 1 hour

To convert minutes into hours:

Divide the number of minutes by 60.

Formula

Hours = Minutes $\div$ 60

Example 1

Convert:

180 minutes into hours

$$180 \div 60 = 3 \text{ hours}$$

Therefore:

180 minutes = 3 hours

Example 2

Convert:

200 minutes into hours and minutes

$$200 \div 60 = 3 \text{ remainder } 20$$

Therefore:

200 minutes = 3 hours 20 minutes.

Example 3

Convert:

100 minutes

$$60 \text{ minutes} = 1 \text{ hour}$$

$$100 - 60 = 40 \text{ minutes}$$

Therefore:

100 minutes = 1 hour 40 minutes

Revision Exercise 2

Convert into hours:

1. 120 minutes
2. 360 minutes
3. 840 minutes
4. 180 minutes
5. 480 minutes

Convert into hours and minutes:

6. 72 minutes
 7. 130 minutes
 8. 90 minutes
 9. 425 minutes
 10. 250 minutes
-

8.3 Converting Hours into Seconds

We know:

1 hour = 60 minutes

and:

1 minute = 60 seconds

Therefore:

1 hour = 60×60

1 hour = 3,600 seconds.

Formula

Seconds = Hours $\times$ 3,600

Example 1

Convert:

3 hours into seconds

$$3 \times 3,600 = \mathbf{10,800 \text{ seconds}}$$

Example 2

Convert:

6 hours 50 seconds

6 hours:

$$6 \times 3,600 = 21,600 \text{ seconds}$$

Add 50 seconds:

$$21,600 + 50 = \mathbf{21,650 \text{ seconds}}$$

Therefore:

$$\mathbf{6 \text{ h } 50 \text{ s} = 21,650 \text{ seconds.}}$$

Converting Minutes into Seconds

Since:

$$\mathbf{1 \text{ minute} = 60 \text{ seconds}}$$

Example

Convert:

10 minutes

$$10 \times 60 = \mathbf{600 \text{ seconds}}$$

Revision Exercise 3

Convert into seconds:

1. 2 hours
 2. 5 hours
 3. 3 hours
 4. 12 hours
 5. 1½ hours
 6. 10 minutes
 7. 20 minutes
 8. 2 hours 30 seconds
-

8.4 Changing Days into Hours

There are:

24 hours in one day

Therefore:

1 day = 24 hours

To convert days into hours:

Multiply the number of days by 24.

Formula

Hours = Days × 24

Example 1

Convert:

5 days into hours

$5 \times 24 = 120$ hours

Therefore:

5 days = 120 hours

Example 2

Convert:

10½ days into hours

10 days:

$$10 \times 24 = 240 \text{ hours}$$

½ day:

$$\frac{1}{2} \times 24 = 12 \text{ hours}$$

Total:

$$240 + 12 = \mathbf{252 \text{ hours}}$$

Therefore:

$$\mathbf{10\frac{1}{2} \text{ days} = 252 \text{ hours.}}$$

Revision Exercise 4

Convert into hours:

1. 10 days
 2. 15 days
 3. 6 days
 4. 13 days
 5. 11 days
 6. 14 days
 7. 5½ days
 8. 2½ days
-

8.5 Changing Hours into Days

Since:

$$\mathbf{24\ hours = 1\ day}$$

To convert hours into days:

Divide the number of hours by 24.

Formula

$$\mathbf{Days = Hours \div 24}$$

Example 1

Convert:

144 hours into days

$$144 \div 24 = \mathbf{6\ days}$$

Therefore:

$$\mathbf{144\ hours = 6\ days}$$

Example 2

Convert:

182 hours into days and hours

$$182 \div 24$$

$$24 \times 7 = 168$$

Remainder:

$$182 - 168 = 14$$

Therefore:

182 hours = 7 days 14 hours.

Revision Exercise 5

Convert into days:

1. 120 hours
2. 240 hours
3. 720 hours
4. 48 hours
5. 96 hours

Convert into days and hours:

6. 612 hours
 7. 312 hours
 8. 249 hours
 9. 200 hours
 10. 150 hours
-

8.6 Finding Time Intervals

The **duration** of an event is the length of time from when it starts to when it ends.

The basic rule is:

Duration = Ending time – Starting time.

For a journey:

- starting time may be called **departure time**,
 - ending time may be called **arrival time**.
-

Example 1

A doctor works from:

8:00 a.m. to 12:00 noon

Duration:

$$12:00 - 8:00 = \mathbf{4 \text{ hours}}$$

Therefore:

The doctor worked for 4 hours.

Example 2

A baby sleeps at:

8:15 a.m.

and wakes at:

10:15 a.m.

Duration:

$$10:15 - 8:15 = \mathbf{2 \text{ hours}}$$

Example 3

A train leaves at:

10:45 a.m.

and arrives at:

12:45 p.m.

Duration:

$$12:45 - 10:45 = \mathbf{2 \text{ hours}}$$

Finding Intervals with Different Minutes

Example

Find the time from:

7:30 a.m. to 11:45 a.m.

From 7:30 to 11:30 = 4 hours

From 11:30 to 11:45 = 15 minutes

Therefore:

Duration = 4 hours 15 minutes

Revision Exercise 6

Find the duration:

1. 9:00 a.m. to 3:00 p.m.
 2. 4:00 a.m. to 1:00 p.m.
 3. 6:00 a.m. to 4:00 p.m.
 4. 8:15 a.m. to 10:15 a.m.
 5. 10:45 a.m. to 12:45 p.m.
 6. 7:30 a.m. to 11:45 a.m.
 7. 1:15 p.m. to 4:50 p.m.
 8. 8:40 a.m. to 9:55 a.m.
-

8.7 Addition of Time

Time quantities can be added.

Remember:

60 minutes = 1 hour

If the total number of minutes reaches 60 or more, convert the extra minutes into hours.

Example 1

A train travels:

7 h 55 min

and then another:

2 h 15 min

Add the hours:

$$7 \text{ h} + 2 \text{ h} = 9 \text{ h}$$

Add the minutes:

$$55 \text{ min} + 15 \text{ min} = 70 \text{ min}$$

$$70 \text{ min} = \mathbf{1 \text{ h } 10 \text{ min}}$$

Therefore:

$$9 \text{ h} + 1 \text{ h } 10 \text{ min}$$

$$= \mathbf{10 \text{ h } 10 \text{ min.}}$$

Example 2

A driver takes:

1 h 20 min

to reach one town.

He then takes:

1 h 45 min

to reach another town.

Add:

$$1 \text{ h} + 1 \text{ h} = 2 \text{ h}$$

$$20 \text{ min} + 45 \text{ min} = 65 \text{ min}$$

$$65 \text{ min} = 1 \text{ h } 5 \text{ min}$$

Therefore:

$$\text{Total time} = 3 \text{ h } 5 \text{ min}$$

8.8 Subtraction of Time

Time can also be subtracted to find:

- a starting time,
 - an ending time,
 - the amount of time remaining.
-

Example

An aircraft arrived at:

12:40 a.m. Tuesday

The journey lasted:

4 h 45 min

Find the departure time.

Subtract 4 hours:

$$12:40 \text{ a.m.} - 4 \text{ h} = 8:40 \text{ p.m. Monday}$$

Subtract 45 minutes:

8:40 p.m. – 45 min = **7:55 p.m. Monday**

Therefore:

The aircraft departed at 7:55 p.m. on Monday.

Finding Ending Time

Example

Lunch starts at:

12:45 p.m.

Lunch lasts:

1 h 15 min

Add 1 hour:

12:45 p.m. + 1 h = 1:45 p.m.

Add 15 minutes:

1:45 p.m. + 15 min = **2:00 p.m.**

Therefore:

Lunch ends at 2:00 p.m.

Real-Life Time Problems

Example 1

A car travels from Town A to Town B in 2 hours.

It then travels from Town B to Town C in 3 hours.

Total duration:

$$2 + 3 = \mathbf{5 \text{ hours}}$$

Example 2

A race ends at:

12:35 p.m.

The race lasts:

2 h 10 min

Subtract:

$$12:35 \text{ p.m.} - 2 \text{ h} = 10:35 \text{ a.m.}$$

$$10:35 \text{ a.m.} - 10 \text{ min} = \mathbf{10:25 \text{ a.m.}}$$

Therefore:

The race began at 10:25 a.m.

IMPORTANT TIME RULES

Hours to Minutes

Multiply by 60

Example:

$$3 \text{ h} = 3 \times 60 = \mathbf{180 \text{ min}}$$

Minutes to Hours

Divide by 60

Example:

$$180 \text{ min} \div 60 = \mathbf{3 \text{ h}}$$

Hours to Seconds

Multiply by 3,600

Example:

$$2 \text{ h} = 7,200 \text{ s}$$

Minutes to Seconds

Multiply by 60

Example:

$$5 \text{ min} = 300 \text{ s}$$

Days to Hours

Multiply by 24

Example:

$$5 \text{ days} = 120 \text{ h}$$

Hours to Days

Divide by 24

Example:

$$72 \text{ h} = 3 \text{ days}$$

Time Interval

Duration = Ending time – Starting time

UNIT 8 REVISION SUMMARY

Remember:

- 1 minute = 60 seconds.
 - 1 hour = 60 minutes.
 - 1 hour = 3,600 seconds.
 - 1 day = 24 hours.
 - Multiply by 60 to change hours to minutes.
 - Divide by 60 to change minutes to hours.
 - Multiply by 24 to change days to hours.
 - Divide by 24 to change hours to days.
 - Duration is the difference between ending time and starting time.
 - When adding minutes, convert every 60 minutes into 1 hour.
 - Time management helps us to be punctual and organised.
-

END-OF-UNIT REVISION EXERCISE

A. Conversion of Time

1. Convert **5 hours** into minutes.
 2. Convert **$\frac{1}{2}$ hour** into minutes.
 3. Convert **150 minutes** into hours and minutes.
 4. Convert **90 minutes** into hours and minutes.
 5. Convert **3 hours** into seconds.
 6. Convert **10 minutes** into seconds.
 7. Convert **7 days** into hours.
 8. Convert **168 hours** into days.
-

B. Time Intervals

9. Find the interval from:

7:30 a.m. to 11:45 a.m.

10. Find the interval from:

1:15 p.m. to 4:50 p.m.

11. An activity begins at **8:40 a.m.** and ends at **9:55 a.m.** Find its duration.

12. A baby sleeps from **8:15 a.m. to 10:15 a.m.** How long does the baby sleep?

C. Addition and Subtraction of Time

13. Calculate:

14 h 18 min + 12 h 32 min

Express the answer in days, hours and minutes.

14. Calculate:

6 days 12 h – 3 days 9 h

15. Add:

1 h 20 min + 1 h 45 min

16. Lunch starts at **12:45 p.m.** and lasts **1 h 15 min.** At what time does lunch end?

17. A race ends at **12:35 p.m.** and lasts **2 h 10 min.** At what time did it begin?

D. Word Problems

18. A bus leaves Kigali at **6:45 a.m.** and arrives in Musanze at **10:20 a.m.** Find the total journey time.

19. If the same bus in Question 18 stops for **5 minutes**, find the actual travelling time.

20. A train takes **7 h 55 min** for the first part of a journey and **2 h 15 min** for the second part. Find the total journey time.

21. A festival lasts **5½ days**. How many hours does it last?

22. A community activity lasts **425 minutes**. Express the time in hours and minutes.
23. A camp lasts **312 hours**. Express the duration in days.
24. Explain why good time management and punctuality are important in school and daily life.

UNIT 9: MONEY AND ITS FINANCIAL APPLICATIONS

Key Unit Competence

By the end of this unit, you should be able to:

- understand the use of money,
 - change Rwandan coins and notes,
 - calculate buying price and selling price,
 - calculate profit and loss,
 - make simple budgets,
 - use money responsibly,
 - convert currencies using a given exchange rate.
-

9.0 Introduction

Money is used in many daily activities such as:

- buying goods,
- paying for services,
- saving,
- budgeting,
- planning expenses,
- running businesses.

This unit introduces changing money, buying and selling, budgeting, profit, loss and currency conversion.

9.1 Rwandan Currency

Money is a **medium of exchange**.

Rwandan money is expressed in **Rwandan francs (Frw)**.

The source lists coins such as:

1 Frw, 5 Frw, 10 Frw, 20 Frw, 50 Frw and 100 Frw

and notes such as:

500 Frw, 1,000 Frw, 2,000 Frw and 5,000 Frw.

Changing Money

Changing money means replacing one amount with other notes or coins whose total value is the same.

Example 1

Change:

5,000 Frw

Possible combinations include:

$5 \times 1,000 \text{ Frw} = 5,000 \text{ Frw}$

or:

$10 \times 500 \text{ Frw} = 5,000 \text{ Frw}$

or:

$2 \times 2,000 \text{ Frw} + 1 \times 1,000 \text{ Frw} = 5,000 \text{ Frw}$

Example 2

Make **100 Frw**.

Possible ways:

1 × 100 Frw

2 × 50 Frw

10 × 10 Frw

The important rule is:

The total value of the notes and coins must equal the original amount.

Finding Change

The basic formula is:

Change = Money given – Cost of goods

Example

A snack costs **350 Frw**.

You pay with **500 Frw**.

Change:

500 – 350 = 150 Frw

Another Example

A book costs **1,200 Frw**.

You pay with **2,000 Frw**.

Change:

$$2,000 - 1,200 = 800 \text{ Frw}$$

Revision Exercise 1

1. How many 500 Frw notes make 1,000 Frw?
 2. How many 100 Frw coins make 2,000 Frw?
 3. Give two ways of making 250 Frw.
 4. A pupil buys juice for 600 Frw and pays 1,000 Frw. Find the change.
 5. A pupil has one 5,000 Frw note, two 1,000 Frw notes and five 100 Frw coins. Find the total amount.
-

9.2 Buying Price and Selling Price

Cost Price or Buying Price

The **Cost Price (CP)** or **Buying Price** is the amount paid by a seller to obtain an item.

Selling Price

The **Selling Price (SP)** is the amount at which the item is sold to the customer.

Profit

A **profit** occurs when:

Selling Price > Cost Price

Formula:

Profit = Selling Price – Cost Price

Loss

A **loss** occurs when:

Cost Price > Selling Price

Formula:

$$\text{Loss} = \text{Cost Price} - \text{Selling Price}.$$

9.2.1 Finding the Cost Price

If there is a profit:

$$\text{CP} = \text{SP} - \text{Profit}$$

If there is a loss:

$$\text{CP} = \text{SP} + \text{Loss}$$

Example 1

A radio is sold for:

50,000 Frw

Profit:

1,300 Frw

Cost Price:

$$50,000 - 1,300 = \mathbf{48,700 \text{ Frw}}$$

Example 2

100 kg of potatoes are sold at:

120 Frw per kg

Selling Price:

$$100 \times 120 = \mathbf{12,000 \text{ Frw}}$$

There is a loss of:

1,000 Frw

Cost Price:

$12,000 + 1,000 = \mathbf{13,000 \text{ Frw.}}$

Revision Exercise 2

1. A bottle is sold for 4,000 Frw with a profit of 800 Frw. Find the cost price.
 2. A piglet is sold for 30,000 Frw with a loss of 3,000 Frw. Find the cost price.
 3. A chair is sold for 25,000 Frw with a profit of 5,000 Frw. Find the buying price.
-

9.2.2 Finding the Selling Price

If there is a profit:

$SP = CP + \text{Profit}$

If there is a loss:

$SP = CP - \text{Loss.}$

Example 1

A cooker costs:

80,000 Frw

Profit:

14,700 Frw

Selling Price:

$80,000 + 14,700 = \mathbf{94,700 \text{ Frw}}$

Example 2

A calculator costs:

9,500 Frw

Loss:

2,200 Frw

Selling Price:

$9,500 - 2,200 = 7,300$ **Frw.**

Revision Exercise 3

1. A bicycle costs 38,000 Frw. The seller makes a profit of 4,000 Frw. Find the selling price.
 2. A lamp costs 3,500 Frw. The seller makes a loss of 500 Frw. Find the selling price.
 3. A bag costs 12,000 Frw and is sold at a profit of 2,500 Frw. Find the selling price.
-

9.2.3 Calculating Profit and Loss

Example 1: Profit

A trader buys a radio at:

13,000 Frw

and sells it at:

15,000 Frw

Profit:

$15,000 - 13,000$

= **2,000 Frw**

Example 2: Loss

A shopkeeper buys shoes for:

21,000 Frw

and sells them for:

17,500 Frw

Loss:

$21,000 - 17,500$

= **3,500 Frw.**

Important Profit and Loss Rules

$SP > CP \rightarrow \text{Profit}$

$CP > SP \rightarrow \text{Loss}$

$SP = CP \rightarrow \text{Neither profit nor loss}$

Revision Exercise 4

Calculate the profit or loss:

1. $CP = 80,000 \text{ Frw}$, $SP = 60,000 \text{ Frw}$
 2. $CP = 6,000 \text{ Frw}$, $SP = 5,000 \text{ Frw}$
 3. $CP = 10,000 \text{ Frw}$, $SP = 12,000 \text{ Frw}$
 4. A shirt costs 4,500 Frw and is sold for 3,800 Frw.
 5. A book costs 5,000 Frw and is sold for 7,500 Frw.
-

9.3 Simple Budgeting

A **budget** is a plan showing how money will be spent.

Budgeting helps us know:

- how much money we have,
 - what we need,
 - what we want,
 - what should be bought first,
 - how to avoid waste.
-

Needs and Wants

Needs

Needs are things that are necessary or important for daily life.

Examples include:

- food,
- shelter,
- clothes,
- water,
- school fees,
- education.

Wants

Wants are things we may desire but can live without.

Examples may include:

- toys,
- video games,
- ice cream,
- television,
- entertainment.

Money should normally be spent on important needs before less important wants.

Example of a Budget

A family has:

50,000 Frw

Planned expenses:

- Food = 20,000 Frw
- Clothes = 12,000 Frw
- School fees = 16,000 Frw
- Painting house = 12,000 Frw

Total required:

$20,000 + 12,000 + 16,000 + 12,000$

= 60,000 Frw

But the family has only:

50,000 Frw

Therefore, the family cannot do everything immediately.

It should spend according to priority and postpone a less urgent expense.

Revision Exercise 5

1. Define a budget.
 2. Give three examples of needs.
 3. Give three examples of wants.
 4. A family has 100,000 Frw but wants to spend 125,000 Frw. How much extra money would be required?
 5. Explain why needs should usually come before wants.
-

9.4 Buying and Selling

Buying and selling can be practised using real-life calculations.

Example

A pupil has:

5,000 Frw

The pupil buys:

- Mathematical set = 1,000 Frw
- Ruler = 500 Frw
- Pen = 200 Frw

Total cost:

$$1,000 + 500 + 200$$

$$= \mathbf{1,700 \text{ Frw}}$$

Change:

$$5,000 - 1,700$$

$$= \mathbf{3,300 \text{ Frw.}}$$

Revision Exercise 6

1. John has 6,000 Frw. He buys four bottles of juice at 1,000 Frw each and one yoghurt for 1,000 Frw. Find his change.
 2. One banana costs 100 Frw. How many bananas can be bought with 1,200 Frw?
 3. A pupil has 10,000 Frw and buys goods costing 7,250 Frw. Find the change.
-

9.5 Converting Currencies

Different countries use different currencies.

To convert money from one currency to another, an **exchange rate** is required.

The source gives an example exchange table from a particular forex shop and time, but it also states that current accepted exchange rates should be used when converting currencies.

For revision, what matters is the **method**.

General Currency Conversion Method

If:

1 unit of foreign currency = R Frw

then:

Frw to Foreign Currency

Foreign currency = Frw amount $\div$ R

Foreign Currency to Frw

Frw = Foreign currency $\times$ R

Example

Suppose:

1 USD = 1,400 Frw

Convert:

14,000 Frw to USD

$14,000 \div 1,400 = 10$ USD

Another Example

Using the same sample rate:

Convert:

20 USD to Frw

$$20 \times 1,400$$

$$= \mathbf{28,000 \text{ Frw}}$$

UNIT 9 IMPORTANT FORMULAS

$$\mathbf{Change = Money \text{ given} - Cost}$$

$$\mathbf{Profit = SP - CP}$$

$$\mathbf{Loss = CP - SP}$$

$$\mathbf{SP = CP + Profit}$$

$$\mathbf{SP = CP - Loss}$$

$$\mathbf{CP = SP - Profit}$$

$$\mathbf{CP = SP + Loss}$$

UNIT 9 REVISION SUMMARY

Remember:

- Money is used as a medium of exchange.
- Notes and coins can be exchanged for other combinations of equal value.
- Cost Price is what a seller pays to obtain an item.
- Selling Price is what the customer pays.

- Profit occurs when SP is greater than CP.
 - Loss occurs when CP is greater than SP.
 - A budget is a plan for spending money.
 - Needs should normally be prioritised before wants.
 - Currency conversion requires an exchange rate.
 - Accuracy and honesty are important when handling money.
-

END-OF-UNIT REVISION EXERCISE

A. Money and Change

1. Give three different ways of changing 5,000 Frw.
 2. A customer pays 5,000 Frw for goods costing 3,450 Frw. Find the change.
 3. How many 500 Frw notes make 4,000 Frw?
-

B. Profit and Loss

4. CP = 15,000 Frw and SP = 19,000 Frw. Find the profit.
 5. CP = 25,000 Frw and SP = 21,500 Frw. Find the loss.
 6. An item sells for 50,000 Frw with a profit of 8,000 Frw. Find the CP.
 7. An item costs 75,000 Frw and is sold at a loss of 5,000 Frw. Find the SP.
 8. A cow is bought for 300,000 Frw and sold for 400,000 Frw. Find the profit.
-

C. Budgeting

9. Explain the difference between a need and a want.
10. A family has 200,000 Frw. It plans to spend:

- Food = 70,000 Frw
- School fees = 80,000 Frw
- Clothes = 30,000 Frw
- Television = 100,000 Frw

Can the family afford everything? Explain.

11. Which expense in Question 10 would you consider postponing first? Explain.

D. Buying and Selling

12. Manzi has 5,000 Frw. He buys:

- Mathematical set = 1,000 Frw
- Ruler = 500 Frw
- Pen = 150 Frw

Find his change.

13. A pupil buys 5 notebooks at 400 Frw each. Find the total cost.

E. Currency Conversion

14. Using an exchange rate of **1 USD = 1,500 Frw**, convert 15,000 Frw to USD.

15. Using the same rate, convert 25 USD to Rwandan francs.

16. Explain why the current exchange rate must be checked before converting currencies.

UNIT 10: SEQUENCES THAT INCLUDE WHOLE NUMBERS, FRACTIONS AND DECIMALS

Key Unit Competence

By the end of this unit, you should be able to:

- arrange whole numbers in increasing and decreasing order,
 - explain sequences involving whole numbers, fractions and decimals,
 - identify patterns,
 - find missing numbers,
 - complete sequences with a constant difference,
 - complete sequences with a constant ratio.
-

10.0 Introduction

A **sequence** is an ordered list of numbers, shapes or objects that follows a particular rule or pattern.

Sequences may involve:

- whole numbers,
- fractions,
- decimals,
- repeated addition,
- repeated subtraction,
- repeated multiplication.

Understanding sequences helps us recognise patterns and predict what comes next.

10.1 Increasing Order

Increasing order means arranging numbers from the **smallest to the largest**.

Example

Arrange:

46,295; 45,690; 68,925

Compare the digits starting from the highest place value.

The correct order is:

45,690; 46,295; 68,925.

Another Example

Arrange:

5,000; 4,000; 9,000

in increasing order.

Answer:

4,000; 5,000; 9,000

Revision Exercise 1

Arrange from smallest to largest:

1. 637,045; 705,365; 673,045; 637,450
 2. 491,279; 137,004; 397,080; 491,792
 3. 2,300; 3,200; 2,003; 3,002
 4. 5,732; 7,532; 735; 5,372
-

10.2 Decreasing Order

Decreasing order means arranging numbers from the **greatest to the smallest**.

Example

Arrange:

43,250; 42,420; 43,502; 40,352

in decreasing order.

Answer:

43,502; 43,250; 42,420; 40,352.

Revision Exercise 2

Arrange in decreasing order:

1. 213,456; 213,564; 213,546; 213,645
 2. 23,451; 23,514; 23,145; 23,415
 3. 860,720; 806,720; 860,270; 860,027
 4. 9,481; 9,841; 9,148; 9,099
-

10.3 Simple Sequences

A sequence follows a particular rule.

To complete a sequence:

1. Identify the rule.
 2. Apply the rule to find the next term.
 3. Check whether the rule fits the previous terms.
-

Example 1: Addition Pattern

3, 6, 9, 12, ...

Rule:

Add 3

Next term:

$$12 + 3 = \mathbf{15}$$

Example 2: Multiplication Pattern

2, 4, 8, 16, ...

Rule:

Multiply by 2

Next term:

$$16 \times 2 = \mathbf{32}$$

Example 3: Square Numbers

1, 4, 9, 16, 25, ...

These are:

$$1^2, 2^2, 3^2, 4^2, 5^2$$

The next term is:

$$6^2 = \mathbf{36}.$$

Revision Exercise 3

Find the next two terms:

1. 10, 15, 20, 25, ____, ____
 2. 7, 14, 21, 28, ____, ____
 3. 15, 26, 37, 48, 59, ____, ____
 4. 2,109; 2,210; 2,311; 2,412; ____; ____
-

10.4 Sequences Involving Fractions

Fraction sequences may increase or decrease by the same fractional amount.

Example

20; 21½; 23; 24½; 26; 27½; ...

Find the difference:

$$21\frac{1}{2} - 20 = 1\frac{1}{2}$$

$$23 - 21\frac{1}{2} = 1\frac{1}{2}$$

Therefore, the rule is:

Add $1\frac{1}{2}$

Next term:

$$27\frac{1}{2} + 1\frac{1}{2} = \mathbf{29}.$$

Example 2

15; $16\frac{1}{2}$; 18; $19\frac{1}{2}$; 21; ...

Rule:

Add $1\frac{1}{2}$

Next terms:

$22\frac{1}{2}$; 24

Revision Exercise 4

Find the next terms:

1. 15; $16\frac{1}{2}$; 18; $19\frac{1}{2}$; 21; ____; ____
 2. 36; $34\frac{1}{2}$; 33; $31\frac{1}{2}$; 30; ____; ____
 3. 50; $52\frac{1}{2}$; 55; $57\frac{1}{2}$; 60; ____; ____
 4. $25\frac{1}{4}$; 25; $24\frac{3}{4}$; $24\frac{1}{2}$; $24\frac{1}{4}$; ____
-

10.5 Sequences Involving Decimals

A decimal sequence may increase or decrease by adding or subtracting a constant decimal value.

Example

5; 5.5; 6; 6.5; 7; 7.5; ...

Difference:

0.5

Therefore, the rule is:

Add 0.5

Next terms:

$$7.5 + 0.5 = 8$$

$$8 + 0.5 = 8.5$$

So:

5; 5.5; 6; 6.5; 7; 7.5; 8; 8.5.

Revision Exercise 5

Find the next terms:

1. 1; 1.5; 2; 2.5; 3; 3.5; 4; ____; ____
 2. 14.5; 13; 11.5; 10; 8.5; 7; ____; ____
 3. 70; 73.5; 77; 80.5; 84; 87.5; ____
 4. 19; 18.5; 18; 17.5; 17; 16.5; 16; ____
-

10.6 Sequences with Constant Difference

A sequence with a constant difference is called an **arithmetic progression**.

The same amount is added or subtracted each time.

Example

2, 4, 6, 8, ...

Differences:

$$4 - 2 = 2$$

$$6 - 4 = 2$$

$$8 - 6 = 2$$

Common difference:

2

Next term:

$$8 + 2 = \mathbf{10}.$$

Example 2

25, 28, 31, 34, 37, ...

Difference:

3

Next term:

$$37 + 3 = \mathbf{40}$$

Revision Exercise 6

Find the missing terms:

1. 35, 37, 39, 41, 43, ____, ____
 2. 7, 11, 15, 19, 23, ____, ____
 3. 40, 43, 46, 49, ____, ____
 4. 60, 64, 68, 72, ____
-

10.7 Sequences with Constant Ratio

A sequence has a **constant ratio** when every term after the first is obtained by multiplying the previous term by the same fixed number.

This is also called a **geometric sequence** or **geometric progression**.

Example 1

1, 2, 4, 8, ...

Each term is multiplied by 2.

Rule:

$\times 2$

Next terms:

$$8 \times 2 = 16$$

$$16 \times 2 = 32$$

Therefore:

1, 2, 4, 8, 16, 32

Example 2

5, 15, 45, ...

Find the ratio:

$$15 \div 5 = 3$$

$$45 \div 15 = 3$$

Common ratio:

3

Next term:

$$45 \times 3 = 135.$$

Example 3: Decreasing Geometric Sequence

176, 88, 44, 22, ...

Each term is divided by 2.

Next terms:

$$22 \div 2 = 11$$

$$11 \div 2 = 5.5$$

Revision Exercise 7

Find the next terms:

1. 1, 2, 4, 8, __, __
 2. 16, 32, 64, __
 3. 1, 5, 25, 125, __
 4. 10, 100, 1,000, __
 5. 176, 88, 44, 22, __, __
-

10.8 Sequences Where the Difference Changes Geometrically

Some sequences do not have a constant difference.

Instead, the **differences themselves follow a geometric sequence**.

Example

Consider:

10, 21, 43, 87, ...

Find the differences:

$$21 - 10 = 11$$

$$43 - 21 = 22$$

$$87 - 43 = 44$$

The differences are:

11, 22, 44

Each difference is multiplied by 2.

Next difference:

$$44 \times 2 = \mathbf{88}$$

Next term:

$$87 + 88 = \mathbf{175}$$

Therefore:

10, 21, 43, 87, 175.

Another Example

1, 3, 7, 15, ...

Differences:

2, 4, 8

Next difference:

16

Next term:

$$15 + 16 = \mathbf{31}$$

Revision Exercise 8

Find the next term or terms:

1. 1, 3, 7, 15, ____
 2. 2, 5, 11, 23, ____, ____
 3. 5, 11, 23, 47, ____
 4. 8, 17, 35, 71, ____
 5. 20, 41, 83, 167, ____
-

IMPORTANT SEQUENCE RULES

Increasing Order

Arrange from:

smallest $\rightarrow$ largest

Decreasing Order

Arrange from:

largest $\rightarrow$ smallest

Arithmetic Sequence

Same number is added or subtracted each time.

Example:

4, 7, 10, 13

Common difference = 3

Geometric Sequence

Each term is multiplied by the same number.

Example:

3, 6, 12, 24

Common ratio = 2

UNIT 10 REVISION SUMMARY

Remember:

- A sequence is an ordered list that follows a rule.
 - Increasing order goes from smallest to largest.
 - Decreasing order goes from largest to smallest.
 - Whole numbers, fractions and decimals can all form sequences.
 - A constant-difference sequence is an arithmetic progression.
 - A constant-ratio sequence is a geometric progression.
 - Some sequences have differences that themselves form a geometric pattern.
 - Always identify the rule before predicting the next term.
-

END-OF-UNIT REVISION EXERCISE

A. Ordering Numbers

1. Arrange in increasing order:

2,300; 3,200; 2,003; 3,002

2. Arrange in decreasing order:

9,481; 9,841; 9,148; 9,099

B. Whole-Number Sequences

3. Complete:

3, 6, 9, 12, ____

4. Complete:

1, 5, 9, 13, ____

5. Write the next three terms in a sequence beginning at 4 and increasing by 7.

C. Fraction and Decimal Sequences

6. Complete:

5; $6\frac{1}{2}$; 8; $9\frac{1}{2}$; ____

7. Complete:

10; 9.5; 9; 8.5; ____; ____

8. Complete:

1.5; 2; 2.5; 3; ____

D. Constant Difference

9. Find the common difference and next term:

5, 8, 11, 14, ____

10. Find the next two terms:

20, 25, 30, 35, ____, ____

E. Constant Ratio

11. Complete:

3, 9, 27, 81, ____

12. Complete:

2, 6, 18, 54, ____

13. Complete:

100, 50, 25, 12.5, ____

F. Changing Differences

14. Complete:

11, 23, 47, 95, ____

15. Complete:

2, 5, 11, 23, ____

16. Complete:

8, 17, 35, 71, ____

G. Word Problem

17. A farmer plants trees in rows:

Row 1 = 3 trees

Row 2 = 6 trees

Row 3 = 9 trees

(a) How many trees will be in Row 5?

(b) Describe the pattern used.

UNIT 12: INTERPRETING AND CONSTRUCTING SCALE DRAWINGS

Key Unit Competence

By the end of this unit, you should be able to:

- explain scale drawings,
 - calculate the scale of a map or drawing,
 - find actual distance,
 - find drawing distance,
 - construct simple scale drawings,
 - solve problems involving scale.
-

12.0 Introduction

A **scale drawing** is a drawing that represents a real object or distance using a smaller or larger size.

Large real distances cannot always fit on a sheet of paper, so they are reduced using a scale.

For example:

1 cm may represent 1 km

or the scale may be written as:

1 : 100,000

Scale drawings are used for:

- maps,
- house plans,
- roads,
- classrooms,
- buildings,

- other real objects.
-

12.1 Meaning of a Scale Drawing

A scale drawing represents an object or space in a reduced or enlarged form while keeping the same proportion.

For example:

A road of **12 km** may be represented by a line of **12 cm** on paper.

Example

Suppose:

Drawing length of a desk = **3 cm**

Actual length of desk = **150 cm**

This means:

3 cm represents 150 cm

To find what 1 cm represents:

$$150 \div 3 = 50$$

Therefore:

1 cm represents 50 cm

Scale:

1 : 50.

12.2 Finding the Scale

To find scale:

1. Measure the drawing length.
2. Find the actual length.
3. Convert both to the **same unit**.
4. Divide drawing length by actual length.
5. Simplify the ratio.

The source gives:

$$\text{Scale} = \text{Drawing length} \div \text{Actual length}$$

The scale has **no units**.

Formula

$$\text{Scale} = \text{Drawing length} / \text{Actual length}$$

Usually it is expressed as:

$$1 : n$$

Example 1

Actual path length:

20 m

Drawing length:

5 cm

Convert:

$$20 \text{ m} = 2,000 \text{ cm}$$

Scale:

$$5 : 2,000$$

Divide both by 5:

$$\mathbf{1 : 400}$$

Therefore:

$$\mathbf{\text{Scale} = 1 : 400.}$$

Example 2

Actual length of a desk = **2 m**

Drawing length = **2 cm**

Convert:

$$2 \text{ m} = 200 \text{ cm}$$

Scale:

$$2 : 200$$

$$\mathbf{= 1 : 100}$$

Revision Exercise 1

Find the scale:

1. Actual length = 1.5 m, drawing length = 5 cm
 2. Actual length = 30 cm, drawing length = 10 cm
 3. Actual diameter = 7 cm, drawing diameter = 3.5 cm
 4. Actual road = 40 km, drawing = 10 cm
 5. Actual flagpole = 10 m, drawing = 5 cm
 6. Actual river = 300 km, drawing = 30 cm
-

12.3 Constructing Scale Drawings

To make a scale drawing:

1. Know the actual measurements.
 2. Choose a suitable scale.
 3. Convert actual measurements to drawing measurements.
 4. Draw the object accurately.
-

Example 1

A road section measures:

Length = **50 m**

Width = **5 m**

Scale:

1 : 500

This means:

1 cm represents 500 cm

500 cm = 5 m

Therefore:

1 cm represents 5 m

Drawing length:

$50 \div 5 = \mathbf{10\text{ cm}}$

Drawing width:

$5 \div 5 = \mathbf{1\text{ cm}}$

Therefore, draw a rectangle:

10 cm by 1 cm.

Example 2

A dining hall measures:

40 m by 35 m

Scale:

1 : 1,000

1 cm represents:

1,000 cm = 10 m

Therefore:

Drawing length:

$40 \div 10 = 4 \text{ cm}$

Drawing width:

$35 \div 10 = 3.5 \text{ cm}$

The scale drawing is:

4 cm by 3.5 cm.

Revision Exercise 2

Using scale **1 : 100**, draw:

1. A rectangle measuring 7 m by 3 m.
2. An equilateral triangle with sides of 5 m.
3. A square with sides of 4 m.

Using scale **1 : 500,000**, find drawing lengths for:

4. 15 km
5. 25 km

6. 45 km
 7. 10 km
-

12.4 Finding Actual Distance

To find actual distance:

1. Measure the drawing length.
 2. Interpret the scale.
 3. Multiply the drawing length by the value represented by 1 unit of the drawing.
-

Formula

If the scale is:

1 : n

then:

Actual distance = Drawing length $\times$ n

Remember to convert the answer to a suitable unit.

Example 1

Scale:

1 : 10,000

Drawing length AB:

7 cm

Since:

1 cm represents 10,000 cm

$$10,000 \text{ cm} = 100 \text{ m}$$

Therefore:

7 cm represents:

$$7 \times 100 \text{ m}$$

$$= \mathbf{700 \text{ m}}$$

Example 2

$$AB = 7 \text{ cm}$$

$$BC = 5 \text{ cm}$$

Total drawing distance:

$$7 + 5 = \mathbf{12 \text{ cm}}$$

At scale 1 : 10,000:

$$1 \text{ cm} = 100 \text{ m}$$

Actual distance:

$$12 \times 100$$

$$= \mathbf{1,200 \text{ m}}$$

$$= \mathbf{1.2 \text{ km.}}$$

Revision Exercise 3

Find the actual distance:

1. Drawing = 10 cm, scale = 1 : 2,500
2. Drawing = 5 cm, scale = 1 : 200,000
3. Drawing = 2.5 cm, scale = 1 : 200,000
4. Drawing = 3.2 cm, scale = 1 : 200,000

5. Drawing = 4.9 cm, scale = 1 : 200,000

12.5 Finding Drawing Length

The drawing length is the size used on paper to represent an actual size.

The source gives:

Drawing length = Scale $\times$ Actual size

or:

Drawing length = Actual size $\div$ value represented by one unit.

Formula

If:

1 cm represents x units

then:

Drawing length = Actual distance $\div$ x

Example

Scale:

1 : 200,000

This means:

1 cm represents 200,000 cm

200,000 cm = 2 km

Therefore:

1 cm represents 2 km

Actual distance PQ:

12 km

Drawing length:

$$12 \div 2 = \mathbf{6 \text{ cm}}$$

Actual distance QR:

18 km

Drawing length:

$$18 \div 2 = \mathbf{9 \text{ cm}}$$

Actual distance RS:

10 km

Drawing length:

$$10 \div 2 = \mathbf{5 \text{ cm.}}$$

Another Example

Actual distance:

120 km

Scale:

1 : 300,000

Convert:

$$300,000 \text{ cm} = 3 \text{ km}$$

Therefore:

1 cm represents 3 km

Drawing length:

$$120 \div 3 = \mathbf{40 \text{ cm}}$$

Revision Exercise 4

Find the drawing length:

1. 350 m at scale 1 : 10,000
 2. 820 m at scale 1 : 10,000
 3. 225 m at scale 1 : 10,000
 4. 60 km at scale 1 : 100,000
 5. 225 km at scale 1 : 100,000
 6. 200 km at scale 1 : 100,000
 7. 120 km at scale 1 : 300,000
-

12.6 Scale Drawing Problems

Example 1

A football field is:

90 m long

Scale:

1 : 1,000

1 cm represents:

$$1,000 \text{ cm} = 10 \text{ m}$$

Drawing length:

$$90 \div 10 = \mathbf{9 \text{ cm}}$$

Example 2

A classroom measures:

8 m by 6 m

Scale:

1 : 200

1 cm represents:

$$200 \text{ cm} = 2 \text{ m}$$

Drawing length:

$$8 \div 2 = \mathbf{4 \text{ cm}}$$

Drawing width:

$$6 \div 2 = \mathbf{3 \text{ cm}}$$

Therefore, the drawing dimensions are:

4 cm by 3 cm

Example 3

A river section is:

1,720 m

Drawing length:

17.2 cm

Convert actual distance:

$$1,720 \text{ m} = 172,000 \text{ cm}$$

Scale:

$$17.2 : 172,000$$

Divide both sides by 17.2:

1 : 10,000

IMPORTANT SCALE RULES

Finding Scale

Scale = Drawing length / Actual length

Both measurements must be in the same unit.

Finding Actual Distance

Actual distance = Drawing length $\times$ scale value

Finding Drawing Length

Drawing length = Actual distance $\div$ scale value

Unit Conversion Reminder

Before calculating scale, both drawing and actual measurements must use the same unit.

Remember:

1 m = 100 cm

1 km = 1,000 m

1 km = 100,000 cm

Understanding Scale Notation

Scale 1 : 100

Means:

1 cm on drawing represents 100 cm in real life

Scale 1 : 1,000

Means:

1 cm represents 1,000 cm

or:

1 cm represents 10 m

Scale 1 : 100,000

Means:

1 cm represents 100,000 cm

or:

1 cm represents 1 km

UNIT 12 REVISION SUMMARY

Remember:

- A scale drawing represents a real object proportionally.
- Scale drawings can reduce or enlarge real objects.
- Large objects and distances can be represented on small sheets of paper using scale.
- Scale has no units.
- Drawing and actual measurements must first be converted into the same unit.
- Scale is drawing length divided by actual length.
- Actual distance is found from the drawing length and scale.

- Drawing length is found from the actual distance and scale.
 - Scale drawings are commonly used in maps and building plans.
-

END-OF-UNIT REVISION EXERCISE

A. Basic Concepts

1. What is a scale drawing?
2. Explain what the scale:

1 : 100

means.

3. Give two situations where scale drawings are used.
-

B. Finding Scale

4. A road is 25 km long. On a map it measures 5 cm. Find the scale.
 5. A flagpole is 10 m high. It is represented by 5 cm on a drawing. Find the scale.
 6. A river is 1,720 m long and is represented by 17.2 cm. Find the scale.
-

C. Finding Drawing Length

7. A football field is 90 m long. Find its drawing length at scale 1 : 1,000.
 8. A classroom is 8 m by 6 m. Find its drawing dimensions at scale 1 : 200.
 9. Using scale 1 : 1,000, find the drawing dimensions of a rectangle measuring 40 m by 20 m.
 10. Using scale 1 : 1,000, find the drawing side of a square whose actual side is 50 m.
-

D. Finding Actual Distance

11. Scale = 1 : 25,000 and map distance = 8 cm. Find the actual distance.
12. Scale = 1 : 200,000 and drawing distance = 6 cm. Find the actual distance in kilometres.
13. Scale = 1 : 300,000 and drawing distance = 9 cm. Find the actual distance in kilometres.

E. Construction

14. Use scale 1 : 2,000 to find drawing dimensions for a rectangular field measuring 80 m by 60 m.
 15. Use scale 1 : 30,000 to find the drawing side of a square field measuring 1,200 m on each side.
 16. At scale 1 : 300,000, find the drawing lengths for:
 - a) 21 km
 - b) 27 km
-

F. Multiple Choice

17. A scale drawing is:
 - A. A colourful drawing
 - B. A proportional representation of a real object
 - C. A freehand drawing
 - D. A drawing with no measurements
18. A scale of 1 : 100 means:
 - A. 1 cm represents 1 cm
 - B. 1 cm represents 100 cm
 - C. 100 cm represents 100 cm
 - D. The drawing is 100 times larger

UNIT 12: INTERPRETING AND CONSTRUCTING SCALE DRAWINGS

Key Unit Competence

By the end of this unit, you should be able to:

- explain scale drawings,
 - calculate the scale of a map or drawing,
 - find actual distance,
 - find drawing distance,
 - construct simple scale drawings,
 - solve problems involving scale.
-

12.0 Introduction

A **scale drawing** is a drawing that represents a real object or distance using a smaller or larger size.

Large real distances cannot always fit on a sheet of paper, so they are reduced using a scale.

For example:

1 cm may represent 1 km

or the scale may be written as:

1 : 100,000

Scale drawings are used for:

- maps,
 - house plans,
 - roads,
 - classrooms,
 - buildings,
 - other real objects.
-

12.1 Meaning of a Scale Drawing

A scale drawing represents an object or space in a reduced or enlarged form while keeping the same proportion.

For example:

A road of **12 km** may be represented by a line of **12 cm** on paper.

Example

Suppose:

Drawing length of a desk = **3 cm**

Actual length of desk = **150 cm**

This means:

3 cm represents 150 cm

To find what 1 cm represents:

$$150 \div 3 = 50$$

Therefore:

1 cm represents 50 cm

Scale:

1 : 50.

12.2 Finding the Scale

To find scale:

1. Measure the drawing length.
2. Find the actual length.
3. Convert both to the **same unit**.
4. Divide drawing length by actual length.
5. Simplify the ratio.

The source gives:

Scale = Drawing length $\div$ Actual length

The scale has **no units**.

Formula

Scale = Drawing length / Actual length

Usually it is expressed as:

1 : n

Example 1

Actual path length:

20 m

Drawing length:

5 cm

Convert:

20 m = 2,000 cm

Scale:

5 : 2,000

Divide both by 5:

1 : 400

Therefore:

Scale = 1 : 400.

Example 2

Actual length of a desk = **2 m**

Drawing length = **2 cm**

Convert:

2 m = 200 cm

Scale:

2 : 200

= **1 : 100**

Revision Exercise 1

Find the scale:

1. Actual length = 1.5 m, drawing length = 5 cm
 2. Actual length = 30 cm, drawing length = 10 cm
 3. Actual diameter = 7 cm, drawing diameter = 3.5 cm
 4. Actual road = 40 km, drawing = 10 cm
 5. Actual flagpole = 10 m, drawing = 5 cm
 6. Actual river = 300 km, drawing = 30 cm
-

12.3 Constructing Scale Drawings

To make a scale drawing:

1. Know the actual measurements.
 2. Choose a suitable scale.
 3. Convert actual measurements to drawing measurements.
 4. Draw the object accurately.
-

Example 1

A road section measures:

Length = **50 m**

Width = **5 m**

Scale:

1 : 500

This means:

1 cm represents 500 cm

500 cm = 5 m

Therefore:

1 cm represents 5 m

Drawing length:

$50 \div 5 = \mathbf{10\text{ cm}}$

Drawing width:

$5 \div 5 = \mathbf{1\text{ cm}}$

Therefore, draw a rectangle:

10 cm by 1 cm.

Example 2

A dining hall measures:

40 m by 35 m

Scale:

1 : 1,000

1 cm represents:

$$1,000 \text{ cm} = 10 \text{ m}$$

Therefore:

Drawing length:

$$40 \div 10 = \mathbf{4 \text{ cm}}$$

Drawing width:

$$35 \div 10 = \mathbf{3.5 \text{ cm}}$$

The scale drawing is:

4 cm by 3.5 cm.

Revision Exercise 2

Using scale **1 : 100**, draw:

1. A rectangle measuring 7 m by 3 m.
2. An equilateral triangle with sides of 5 m.
3. A square with sides of 4 m.

Using scale **1 : 500,000**, find drawing lengths for:

4. 15 km
 5. 25 km
 6. 45 km
 7. 10 km
-

12.4 Finding Actual Distance

To find actual distance:

1. Measure the drawing length.
2. Interpret the scale.

3. Multiply the drawing length by the value represented by 1 unit of the drawing.
-

Formula

If the scale is:

1 : n

then:

Actual distance = Drawing length $\times$ n

Remember to convert the answer to a suitable unit.

Example 1

Scale:

1 : 10,000

Drawing length AB:

7 cm

Since:

1 cm represents 10,000 cm

10,000 cm = 100 m

Therefore:

7 cm represents:

$7 \times 100 \text{ m}$

= 700 m

Example 2

$$AB = 7 \text{ cm}$$

$$BC = 5 \text{ cm}$$

Total drawing distance:

$$7 + 5 = \mathbf{12 \text{ cm}}$$

At scale 1 : 10,000:

$$1 \text{ cm} = 100 \text{ m}$$

Actual distance:

$$12 \times 100$$

$$= \mathbf{1,200 \text{ m}}$$

$$= \mathbf{1.2 \text{ km.}}$$

Revision Exercise 3

Find the actual distance:

1. Drawing = 10 cm, scale = 1 : 2,500
 2. Drawing = 5 cm, scale = 1 : 200,000
 3. Drawing = 2.5 cm, scale = 1 : 200,000
 4. Drawing = 3.2 cm, scale = 1 : 200,000
 5. Drawing = 4.9 cm, scale = 1 : 200,000
-

12.5 Finding Drawing Length

The drawing length is the size used on paper to represent an actual size.

The source gives:

Drawing length = Scale $\times$ Actual size

or:

Drawing length = Actual size $\div$ value represented by one unit.

Formula

If:

1 cm represents x units

then:

Drawing length = Actual distance $\div$ x

Example

Scale:

1 : 200,000

This means:

1 cm represents 200,000 cm

200,000 cm = 2 km

Therefore:

1 cm represents 2 km

Actual distance PQ:

12 km

Drawing length:

$12 \div 2 = 6$ cm

Actual distance QR:

18 km

Drawing length:

$$18 \div 2 = \mathbf{9 \text{ cm}}$$

Actual distance RS:

10 km

Drawing length:

$$10 \div 2 = \mathbf{5 \text{ cm.}}$$

Another Example

Actual distance:

120 km

Scale:

1 : 300,000

Convert:

$$300,000 \text{ cm} = 3 \text{ km}$$

Therefore:

1 cm represents 3 km

Drawing length:

$$120 \div 3 = \mathbf{40 \text{ cm}}$$

Revision Exercise 4

Find the drawing length:

1. 350 m at scale 1 : 10,000
 2. 820 m at scale 1 : 10,000
 3. 225 m at scale 1 : 10,000
 4. 60 km at scale 1 : 100,000
 5. 225 km at scale 1 : 100,000
 6. 200 km at scale 1 : 100,000
 7. 120 km at scale 1 : 300,000
-

12.6 Scale Drawing Problems

Example 1

A football field is:

90 m long

Scale:

1 : 1,000

1 cm represents:

1,000 cm = 10 m

Drawing length:

$90 \div 10 = 9$ **cm**

Example 2

A classroom measures:

8 m by 6 m

Scale:

1 : 200

1 cm represents:

$$200 \text{ cm} = 2 \text{ m}$$

Drawing length:

$$8 \div 2 = \mathbf{4 \text{ cm}}$$

Drawing width:

$$6 \div 2 = \mathbf{3 \text{ cm}}$$

Therefore, the drawing dimensions are:

4 cm by 3 cm

Example 3

A river section is:

1,720 m

Drawing length:

17.2 cm

Convert actual distance:

$$1,720 \text{ m} = 172,000 \text{ cm}$$

Scale:

$$17.2 : 172,000$$

Divide both sides by 17.2:

1 : 10,000

IMPORTANT SCALE RULES

Finding Scale

$$\text{Scale} = \text{Drawing length} / \text{Actual length}$$

Both measurements must be in the same unit.

Finding Actual Distance

$$\text{Actual distance} = \text{Drawing length} \times \text{scale value}$$

Finding Drawing Length

$$\text{Drawing length} = \text{Actual distance} \div \text{scale value}$$

Unit Conversion Reminder

Before calculating scale, both drawing and actual measurements must use the same unit.

Remember:

$$1 \text{ m} = 100 \text{ cm}$$

$$1 \text{ km} = 1,000 \text{ m}$$

$$1 \text{ km} = 100,000 \text{ cm}$$

Understanding Scale Notation

Scale 1 : 100

Means:

1 cm on drawing represents 100 cm in real life

Scale 1 : 1,000

Means:

1 cm represents 1,000 cm

or:

1 cm represents 10 m

Scale 1 : 100,000

Means:

1 cm represents 100,000 cm

or:

1 cm represents 1 km

UNIT 12 REVISION SUMMARY

Remember:

- A scale drawing represents a real object proportionally.
 - Scale drawings can reduce or enlarge real objects.
 - Large objects and distances can be represented on small sheets of paper using scale.
 - Scale has no units.
 - Drawing and actual measurements must first be converted into the same unit.
 - Scale is drawing length divided by actual length.
 - Actual distance is found from the drawing length and scale.
 - Drawing length is found from the actual distance and scale.
 - Scale drawings are commonly used in maps and building plans.
-

END-OF-UNIT REVISION EXERCISE

A. Basic Concepts

1. What is a scale drawing?
2. Explain what the scale:

1 : 100

means.

3. Give two situations where scale drawings are used.
-

B. Finding Scale

4. A road is 25 km long. On a map it measures 5 cm. Find the scale.
 5. A flagpole is 10 m high. It is represented by 5 cm on a drawing. Find the scale.
 6. A river is 1,720 m long and is represented by 17.2 cm. Find the scale.
-

C. Finding Drawing Length

7. A football field is 90 m long. Find its drawing length at scale 1 : 1,000.
 8. A classroom is 8 m by 6 m. Find its drawing dimensions at scale 1 : 200.
 9. Using scale 1 : 1,000, find the drawing dimensions of a rectangle measuring 40 m by 20 m.
 10. Using scale 1 : 1,000, find the drawing side of a square whose actual side is 50 m.
-

D. Finding Actual Distance

11. Scale = 1 : 25,000 and map distance = 8 cm. Find the actual distance.
 12. Scale = 1 : 200,000 and drawing distance = 6 cm. Find the actual distance in kilometres.
 13. Scale = 1 : 300,000 and drawing distance = 9 cm. Find the actual distance in kilometres.
-

E. Construction

14. Use scale 1 : 2,000 to find drawing dimensions for a rectangular field measuring 80 m by 60 m.
 15. Use scale 1 : 30,000 to find the drawing side of a square field measuring 1,200 m on each side.
 16. At scale 1 : 300,000, find the drawing lengths for:
 - a) 21 km
 - b) 27 km
-

F. Multiple Choice

17. A scale drawing is:
 - A. A colourful drawing
 - B. A proportional representation of a real object
 - C. A freehand drawing
 - D. A drawing with no measurements
18. A scale of 1 : 100 means:
 - A. 1 cm represents 1 cm
 - B. 1 cm represents 100 cm
 - C. 100 cm represents 100 cm
 - D. The drawing is 100 times larger

UNIT 13: CALCULATING CIRCUMFERENCE OF A CIRCLE, SURFACE AREA AND VOLUME OF CUBES AND CUBOIDS

Key Unit Competence

By the end of this unit, you should be able to:

- explain circumference,
- calculate the circumference of circles,
- understand the relationship between radius, diameter and circumference,

- understand the value of π ,
 - identify properties of cubes and cuboids,
 - calculate surface area,
 - calculate volume,
 - find missing dimensions of cuboids,
 - solve real-life problems involving circles, cubes and cuboids.
-

13.0 Introduction

This unit deals with three important ideas:

- circumference of a circle,
- surface area of cubes and cuboids,
- volume of cubes and cuboids.

These ideas are useful in real life.

For example:

- circumference can help us find the length of fencing needed around a circular garden,
 - surface area can help us find the amount of wrapping paper needed for a box,
 - volume can help us know how much space or liquid a container can hold.
-

13.1 Circumference of Circular Objects

The **circumference** is the total distance around a circle.

It can be measured by:

1. wrapping a string or rope around the circular object,
2. straightening the string,
3. measuring its length.

It can also be calculated using the diameter and π .

Parts of a Circle

Centre

The **centre** is the fixed point in the middle of the circle.

Radius

The **radius** is the distance from the centre to any point on the circumference.

Symbol:

r

Diameter

The **diameter** is a straight line passing through the centre from one side of the circle to the other.

The diameter is twice the radius.

$$\mathbf{D = 2r}$$

Circumference

The circumference is the distance around the circle.

Circumference Formula

The source uses:

$$\mathbf{C = \pi \times D}$$

Since:

$$\mathbf{D = 2r}$$

we can also write:

$$\mathbf{C = 2 \times \pi \times r}$$

Using $\pi \approx 3.14$:

$$C = 3.14 \times D$$

or:

$$C = 2 \times 3.14 \times r.$$

Example 1

Find the circumference of a circle whose diameter is:

14 cm

Formula:

$$C = \pi \times D$$

$$C = 3.14 \times 14$$

$$C = \mathbf{43.96 \text{ cm}}$$

Example 2

Diameter:

20 cm

$$C = 3.14 \times 20$$

$$= \mathbf{62.8 \text{ cm.}}$$

13.2 The Number Pi (π)

Pi, written π , is the constant ratio between the circumference of a circle and its diameter.

Formula:

$$\pi = C \div D$$

The source uses:

$$\pi \approx 3.14$$

and sometimes:

$$\pi \approx 22/7$$

for easier calculation.

Important Relationships

$$\pi = C \div D$$

Therefore:

$$C = \pi D$$

and:

$$D = C \div \pi$$

Also:

$$D = 2r$$

and:

$$r = D \div 2$$

Example 3

A circular garden has radius:

$$7 \text{ m}$$

Diameter:

$$D = 2 \times 7$$

$$= \mathbf{14\ m}$$

Circumference:

$$C = 14 \times 3.14$$

$$= \mathbf{43.96\ m}$$

Therefore, 43.96 m of fencing is needed.

Example 4: Finding Diameter

Circumference:

$$\mathbf{88\ cm}$$

Use:

$$\pi = 22/7$$

Formula:

$$D = C \div \pi$$

$$D = 88 \div 22/7$$

$$D = 88 \times 7/22$$

$$D = \mathbf{28\ cm}$$

Finding Radius

$$r = D \div 2$$

$$r = 28 \div 2$$

$$= \mathbf{14\ cm.}$$

Revision Exercise 1

1. Find the circumference of a circle with diameter 21 cm. Use $\pi = 22/7$.
 2. Find the circumference of a bicycle wheel with diameter 98 cm.
 3. Find the circumference of a circle with radius 15 cm.
 4. A bucket lid has radius 45 cm. Find its circumference.
 5. A circle has circumference 110 m. Find its diameter using $\pi = 22/7$.
 6. A roundabout has circumference 30.8 m. Find its radius using $\pi = 22/7$.
 7. A circle has circumference 439.6 cm. Find its diameter using $\pi = 3.14$.
-

13.3 Properties of Cubes and Cuboids

Cube

A cube has:

- 6 square faces,
- 12 equal edges,
- 8 vertices,
- all faces equal.

Examples include:

- dice,
- sugar cubes,
- some boxes.

Cuboid

A cuboid has:

- 6 rectangular faces,
- 12 edges,
- 8 vertices,
- opposite faces equal.

Its length, width and height may be different.

Cube and Cuboid Comparison

Property	Cube	Cuboid
Faces	6	6
Edges	12	12
Vertices	8	8
Faces	All equal squares Rectangles; opposite faces equal	
Length, width, height	Equal	May be different

Revision Exercise 2

1. How many faces does a cube have?
 2. How many edges does a cuboid have?
 3. How many vertices does a cube have?
 4. State one difference between a cube and a cuboid.
 5. Give two examples of cube-shaped objects.
 6. Give two examples of cuboid-shaped objects.
-

13.4 Surface Area of a Cube

A **net** is the flat shape obtained when a solid is opened out.

A cube has six identical square faces.

Therefore:

$$\text{Surface Area} = \text{Area of one face} \times 6$$

If the side length is **a**:

$$\text{SA} = 6 \times a \times a$$

or:

$$SA = 6a^2.$$

Example

Side of cube:

$$\mathbf{5\text{ cm}}$$

Surface area:

$$SA = 6 \times 5 \times 5$$

$$= 6 \times 25$$

$$= \mathbf{150\text{ cm}^2}.$$

Finding the Side from Surface Area

Suppose:

$$\text{Surface area} = \mathbf{96\text{ cm}^2}$$

Formula:

$$6 \times \text{side} \times \text{side} = 96$$

Divide by 6:

$$\text{side} \times \text{side} = 16$$

Since:

$$4 \times 4 = 16$$

$$\text{Side} = \mathbf{4\text{ cm}}.$$

Revision Exercise 3

1. Find the surface area of a cube of side 3 cm.
 2. Find the surface area of a cube of side 7 cm.
 3. A cube has surface area 216 cm^2 . Find its side length.
 4. A gift box is shaped like a cube with sides of 3 cm. How much paper is needed to cover it completely?
-

13.5 Surface Area of a Cuboid

A cuboid has three pairs of equal rectangular faces.

If:

- L = length
- W = width
- H = height

then:

$$\text{SA} = 2(\text{L} \times \text{W}) + 2(\text{L} \times \text{H}) + 2(\text{W} \times \text{H}).$$

Example

Cuboid dimensions:

$$\mathbf{6 \text{ cm} \times 3 \text{ cm} \times 2 \text{ cm}}$$

Calculate:

$$2(\text{L} \times \text{W})$$

$$= 2(6 \times 3)$$

$$= 36$$

$$2(\text{L} \times \text{H})$$

$$= 2(6 \times 2)$$

$$= 24$$

$$2(W \times H)$$

$$= 2(3 \times 2)$$

$$= 12$$

Total:

$$36 + 24 + 12$$

$$= \mathbf{72 \text{ cm}^2}.$$

Another Example

Cuboid:

$$24 \text{ cm} \times 15 \text{ cm} \times 10 \text{ cm}$$

$$SA = 2(24 \times 15) + 2(15 \times 10) + 2(24 \times 10)$$

$$= 720 + 300 + 480$$

$$= \mathbf{1,500 \text{ cm}^2}.$$

Revision Exercise 4

Find the total surface area:

1. $L = 8 \text{ cm}$, $W = 4 \text{ cm}$, $H = 3 \text{ cm}$
 2. $L = 10 \text{ cm}$, $W = 6 \text{ cm}$, $H = 5 \text{ cm}$
 3. A brick measures $20 \text{ cm} \times 15 \text{ cm} \times 8 \text{ cm}$.
 4. A cuboid measures $12 \text{ cm} \times 9 \text{ cm} \times 4 \text{ cm}$.
-

13.6 Volume of Cubes and Cuboids

Volume is the amount of space occupied by an object.

Volume is measured in **cubic units**, such as:

- cm^3 ,
- m^3 .

For a cuboid:

$$V = L \times W \times H$$

For a cube:

$$V = \text{side} \times \text{side} \times \text{side}$$

or:

$$V = a^3.$$

Volume of a Cube

Example 1

Side:

6 cm

$$V = 6 \times 6 \times 6$$

$$= \mathbf{216 \text{ cm}^3}$$

Example 2

Side:

5 cm

$$V = 5 \times 5 \times 5$$

$$= 125 \text{ cm}^3.$$

Volume of a Cuboid

Example

Dimensions:

$$38 \text{ cm} \times 21 \text{ cm} \times 15 \text{ cm}$$

$$V = 38 \times 21 \times 15$$

$$= 11,970 \text{ cm}^3$$

Another Example

Dimensions:

$$24 \text{ cm} \times 15 \text{ cm} \times 10 \text{ cm}$$

$$V = 24 \times 15 \times 10$$

$$= 3,600 \text{ cm}^3.$$

Revision Exercise 5

1. Find the volume of a cube with sides 15 cm.
 2. Find the volume of a cuboid measuring $32 \text{ cm} \times 18 \text{ cm} \times 18 \text{ cm}$.
 3. A tank measures $4.3 \text{ m} \times 2.4 \text{ m} \times 1.5 \text{ m}$. Find its volume.
 4. A carton measures $64 \text{ cm} \times 32 \text{ cm} \times 30 \text{ cm}$. Find its volume.
 5. Find the volume of a cube with sides 20 cm.
 6. A water tank measures $4 \text{ m} \times 3 \text{ m} \times 2 \text{ m}$. Find its volume.
 7. A brick measures $20 \text{ cm} \times 15 \text{ cm} \times 8 \text{ cm}$. Find its volume.
-

13.7 Finding a Missing Dimension of a Cuboid

Since:

$$V = L \times W \times H$$

we can find any missing dimension by dividing the volume by the product of the other two dimensions.

Formulas

$$L = V \div (W \times H)$$

$$W = V \div (L \times H)$$

$$H = V \div (L \times W).$$

Example 1: Finding Height

Volume:

$$420 \text{ cm}^3$$

Length:

$$10 \text{ cm}$$

Width:

$$7 \text{ cm}$$

Formula:

$$H = V \div (L \times W)$$

$$H = 420 \div (10 \times 7)$$

$$H = 420 \div 70$$

$$= 6 \text{ cm.}$$

Example 2: Finding Width

Volume:

$$4,536 \text{ cm}^3$$

Length:

$$21 \text{ cm}$$

Height:

$$12 \text{ cm}$$

$$W = 4,536 \div (21 \times 12)$$

$$= 4,536 \div 252$$

$$= 18 \text{ cm.}$$

Example 3: Finding Length

Volume:

$$7 \text{ m}^3$$

Width:

$$2 \text{ m}$$

Height:

$$1.4 \text{ m}$$

$$L = 7 \div (2 \times 1.4)$$

$$= 7 \div 2.8$$

$$= 2.5 \text{ m.}$$

Revision Exercise 6

1. $V = 288 \text{ m}^3$, $W = 6 \text{ m}$, $H = 4 \text{ m}$. Find L .
 2. $V = 25,200 \text{ cm}^3$, $L = 36 \text{ cm}$, $W = 50 \text{ cm}$. Find H .
 3. A carton has volume $142,560 \text{ cm}^3$, width 36 cm and height 72 cm . Find its length.
 4. A tank has volume $414,720 \text{ cm}^3$, length 144 cm and width 36 cm . Find its height.
 5. A log has volume $25,200 \text{ cm}^3$, length 35 cm and height 30 cm . Find its width.
-

13.8 Finding Height from Volume and Base Area

For a cuboid:

$$\text{Volume} = \text{Base Area} \times \text{Height}$$

Therefore:

$$V = \text{BSA} \times H$$

and:

$$H = V \div \text{BSA}.$$

Example

Volume:

$$72,000 \text{ cm}^3$$

Base area:

$$2,400 \text{ cm}^2$$

Height:

$$H = 72,000 \div 2,400$$

$$= 30 \text{ cm}.$$

Revision Exercise 7

Find the height:

1. Volume = $3,150 \text{ cm}^3$, base area = 315 cm^2
 2. Volume = 576 cm^3 , base area = 144 cm^2
 3. Volume = 250 m^3 , base area = 50 m^2
 4. Volume = $8,000 \text{ cm}^3$, base area = 160 cm^2
 5. A classroom has base area 56 m^2 and volume 168 m^3 . Find its height.
-

13.9 Finding the Area of a Face of a Cuboid

We know:

$$V = \text{Base Area} \times \text{Height}$$

Therefore:

$$\text{Base Area} = V \div H.$$

Example

Volume:

$$2,618 \text{ cm}^3$$

Height:

$$17 \text{ cm}$$

Base area:

$$2,618 \div 17$$

$$= 154 \text{ cm}^2.$$

Revision Exercise 8

1. Volume = 3,120 cm³, height = 15 cm. Find the base area.
 2. Volume = 1,428 cm³, height = 7 cm. Find the base area.
 3. A box has volume 1.44 m³, length 1.5 m and width 0.8 m. a) Find its base area. b) Find its height.
-

IMPORTANT FORMULAS

Circle

$$D = 2r$$

$$r = D \div 2$$

$$C = \pi D$$

$$C = 2\pi r$$

$$D = C \div \pi$$

Cube

$$\text{Surface Area} = 6a^2$$

$$\text{Volume} = a^3$$

Cuboid

$$\text{Surface Area} = 2(LW) + 2(LH) + 2(WH)$$

$$\text{Volume} = L \times W \times H$$

Missing Dimension

$$L = V \div (W \times H)$$

$$W = V \div (L \times H)$$

$$H = V \div (L \times W)$$

Base Area

$$BSA = V \div H$$

UNIT 13 REVISION SUMMARY

Remember:

- Circumference is the distance around a circle.
 - Diameter is twice the radius.
 - π is approximately 3.14.
 - Circumference can be found using $\pi \times$ diameter.
 - A cube has 6 faces, 12 edges and 8 vertices.
 - A cuboid also has 6 faces, 12 edges and 8 vertices.
 - The surface area of a cube is 6 times the area of one face.
 - The surface area of a cuboid is the sum of the areas of its six faces.
 - Volume is measured in cubic units.
 - Volume of a cube is side $\times$ side $\times$ side.
 - Volume of a cuboid is length $\times$ width $\times$ height.
 - Missing dimensions can be found by rearranging the volume formula.
-

END-OF-UNIT REVISION EXERCISE

A. Circles

1. State the relationship between radius and diameter.
2. Write the formula for the circumference of a circle.

3. Find the circumference of a wheel with diameter 28 cm using $\pi = 22/7$.
 4. A circular garden has radius 14 m. Find its circumference.
 5. A circle has circumference 88 m. Find its diameter using $\pi = 22/7$.
-

B. Cubes

6. A cube has side length 4 cm. Find: a) its surface area, b) its volume.
 7. A cube has side 10 cm. Find its surface area.
 8. A cube has a surface area of 150 cm². Find its side length.
-

C. Cuboids

9. A cuboid measures 8 m $\times$ 5 m $\times$ 3 m. Find: a) total surface area, b) volume.
 10. A brick measures 20 cm $\times$ 15 cm $\times$ 8 cm. Find: a) total surface area, b) volume.
-

D. Missing Dimensions

11. A cuboid has volume 420 cm³, length 10 cm and width 7 cm. Find its height.
 12. A cuboid has volume 4,536 cm³, length 21 cm and height 12 cm. Find its width.
 13. A tank has volume 7 m³, width 2 m and height 1.4 m. Find its length.
 14. A cuboid has volume 8,000 cm³ and base area 160 cm². Find its height.
 15. A cuboid has volume 2,618 cm³ and height 17 cm. Find its base area.
-

E. Word Problems

16. A circular fence has circumference 154 m. Find its diameter using $\pi = 22/7$.
17. A gift box is a cube with sides of 5 cm. How much wrapping paper is required to cover it completely?
18. A rectangular water tank measures 4 m by 3 m by 2 m. How many cubic metres of space does it contain?
19. Explain one real-life use of circumference.
20. Explain one real-life use of surface area and one real-life use of volume.

UNIT 14: DATA HANDLING

Key Unit Competence

By the end of this unit, you should be able to:

- collect data,
 - distinguish discrete and continuous data,
 - record data in tables,
 - use tally charts,
 - represent data using bar charts,
 - represent data using line graphs,
 - interpret graphs to answer questions and solve problems.
-

14.0 Introduction

Data is information that is collected for a particular purpose.

Data helps people:

- know what is happening,
 - remember what happened in the past,
 - compare information,
 - make good plans and decisions.
-

14.1 Quantitative Data

Quantitative data is numerical information that can be counted or measured.

It answers questions such as:

- How many?
- How much?

Examples:

- 25 pupils in a class,
 - a plant that is 15 cm tall,
 - a temperature of 30°C.
-

Qualitative Data

Qualitative data describes qualities, characteristics or categories.

It does not normally answer with numerical measurements.

Examples:

- red flower,
 - spicy soup,
 - dog,
 - favourite hobby.
-

Example

A plant is:

15 cm tall

and has:

fuzzy leaves

Quantitative data:

15 cm

Qualitative data:

fuzzy leaves

Revision Exercise 1

Identify each as quantitative or qualitative:

1. A cat has 4 legs.
 2. The soup is salty.
 3. A race lasted 10 minutes.
 4. The flower is red.
 5. A bag weighs 3 kg.
 6. The shirt is blue.
-

14.2 Types of Quantitative Data

There are two main types:

1. **Discrete data**
 2. **Continuous data**
-

14.2.1 Discrete Data

Discrete data consists of countable, separate values.

It usually uses whole numbers and does not normally include fractions or decimals.

Examples:

- number of pupils,
- number of cars,
- number of goals,
- number of books,
- number of pets.

You can have:

7 apples

but not:

7.5 apples

when counting whole apples.

Revision Exercise 2

Which are discrete data?

1. 8 children
 2. 2.5 kg of rice
 3. 12 books
 4. 3.6 litres of water
 5. 5 puppies
 6. 20 cars
-

14.2.2 Continuous Data

Continuous data is measurable data that can take any value within a range, including decimals and fractions.

Examples include:

- height,
- mass,
- time,
- temperature,
- volume.

Examples:

2.75 m

3.6 kg

12.3 seconds

23.7°C

0.5 litres

Discrete and Continuous Data Comparison

Discrete Data	Continuous Data
Counted	Measured
Usually whole numbers	May include decimals and fractions
Separate values	Can take many values within a range
Example: 12 books	Example: 2.5 kg

Revision Exercise 3

Write **Discrete** or **Continuous**:

1. Number of legs on a chair
 2. Time taken to eat lunch
 3. Pages in a book
 4. Length of a rope
 5. Weight of a baby
 6. Number of eggs in a carton
 7. Temperature of water
 8. Number of goals in a match
-

14.3 Recording Data Using a Tally Chart

A **tally chart** is a simple way of recording and counting data.

The number of times something occurs is called its **frequency**.

How to Make a Tally Chart

1. Write the categories.
 2. Record each response using a tally mark.
 3. Group tally marks in sets of five.
 4. Count the tally marks.
 5. Write the total in the frequency column.
 6. Check that all frequencies add up to the total number of responses.
-

Example

Suppose pupils travel to school using:

- walking,
- bus,
- car,
- motorbike,
- bicycle.

A frequency table may look like:

Transport Frequency

Walk	13
Bus	7
Car	4
Motorbike	5
Bicycle	1
Total	30

The most common method is:

Walking

The least common is:

Bicycle.

Revision Exercise 4

A class survey gives:

- Mathematics = 11 pupils
 - English = 9 pupils
 - SET = 7 pupils
 - Kinyarwanda = 13 pupils
1. Draw a tally chart.
 2. Find the total number of pupils.
 3. Which subject is most preferred?
 4. Which subject is least preferred?
-

14.4 Representing Data Using Bar Charts

A **bar chart** is a graph that uses rectangular bars to represent and compare data.

The height or length of each bar shows the quantity represented.

Main Parts of a Bar Chart

Horizontal Axis

Also called the **x-axis**.

It normally shows categories.

Examples:

- names,
- colours,
- fruits,

- sports.

Vertical Axis

Also called the **y-axis**.

It shows numerical values.

Examples:

- number of pupils,
- number of cars,
- height,
- scores.

Bars

Bars represent the quantities.

A taller bar means a greater quantity.

A shorter bar means a smaller quantity.

Steps for Drawing a Bar Chart

1. Prepare the data.
2. Draw the horizontal and vertical axes.
3. Label both axes.
4. Choose a suitable scale.
5. Draw bars of equal width.
6. Leave equal spaces between the bars.
7. Draw each bar to the correct height.
8. Give the chart a title.

Example

Number of books owned by pupils:

Name	Books
Olivier	10
Thierry	12
Elie	8
Christine	15
Carene	13

The horizontal axis should show the pupils' names.

The vertical axis should show the number of books.

Revision Exercise 5

Represent the following information using a bar chart:

Sport	Pupils
Football	14
Basketball	12
Volleyball	10
Table Tennis	5
Handball	6

Then answer:

1. Which sport is most preferred?
2. Which sport is least preferred?
3. How many pupils were surveyed altogether?

14.5 Interpreting Bar Charts

To interpret a bar chart:

1. Read the title.
 2. Check what is shown on the horizontal axis.
 3. Check what is shown on the vertical axis.
 4. Read the scale.
 5. Compare bar heights.
 6. Identify the greatest and smallest values.
 7. Use the graph to answer questions.
-

Example

Suppose a bar chart shows the heights of family members.

If the father's bar reaches:

180 cm

then:

Father = 180 cm

If the girl's bar reaches:

150 cm

then:

Girl = 150 cm.

Advantages of Bar Charts

Bar charts help us:

- compare quantities quickly,

- identify the highest and lowest values,
 - present data clearly,
 - understand information more easily.
-

Revision Exercise 6

A bar chart represents scores:

- Aline = 15
- James = 12
- Ruth = 18
- Rosy = 10
- Seth = 12

Answer:

1. Who scored highest?
 2. Who scored lowest?
 3. Which pupils got the same score?
 4. How many more marks did Ruth score than Rosy?
-

14.6 Representing Data Using Line Graphs

A **line graph** is used to show how data changes across time or another ordered category.

Points are plotted and joined using straight lines.

Steps for Drawing a Line Graph

1. Collect the data.
2. Choose a suitable scale.
3. Draw and label the axes.
4. Put categories or time on the horizontal axis.
5. Put values on the vertical axis.

6. Plot each point.
 7. Join the points using straight lines.
 8. Add a suitable title.
-

Example

Distance covered by a cyclist:

Time	Distance
8:00 a.m.	0 km
9:00 a.m.	11 km
10:00 a.m.	22 km
11:00 a.m.	33 km
12:00 p.m.	44 km
1:00 p.m.	44 km
2:00 p.m.	55 km

Notice:

From 12:00 p.m. to 1:00 p.m., the distance stays at **44 km**.

This creates a horizontal part of the graph.

Revision Exercise 7

Draw a line graph for:

Time	Distance
10:00 a.m.	0 km

Time	Distance
11:00 a.m.	50 km
12:00 noon	100 km
1:00 p.m.	150 km
2:00 p.m.	150 km
3:00 p.m.	250 km
4:00 p.m.	350 km

Then answer:

1. At what time had 100 km been covered?
 2. During which period did the distance not change?
 3. What was the final distance?
-

14.7 Interpreting Line Graphs

To interpret a line graph:

1. Read the title.
 2. Read the axis labels.
 3. Check the scale.
 4. Follow the plotted points.
 5. Observe whether the line rises, falls or stays level.
 6. Compare the highest and lowest values.
-

Understanding Trends

Rising Line

A rising line means:

The value is increasing.

Falling Line

A falling line means:

The value is decreasing.

Horizontal Line

A horizontal line means:

There is no change.

Example

For an athlete:

At **4 seconds**, the athlete has covered:

25 m

At **8 seconds**:

50 m

At **12 seconds**:

75 m.

Bar Chart vs Line Graph

Bar Chart

Best for categories

Uses bars

Line Graph

Best for changes over time

Uses points and lines

Bar Chart

Useful for comparison

Example: favourite sports

Line Graph

Useful for trends

Example: temperature over a week

14.8 Collecting and Graphing Real-Life Data

A useful data investigation normally follows these steps:

1. Choose a question or event to investigate.
2. Collect data.
3. Organise the data in a tally or frequency table.
4. Choose an appropriate graph.
5. Draw the graph.
6. Interpret the information.
7. Draw a conclusion.

Examples include:

- daily temperature,
- favourite fruits,
- favourite subjects,
- rainfall,
- transport used by pupils.

Bar graphs are suitable for categories, while line graphs are suitable for changes over time.

IMPORTANT DATA HANDLING RULES

Quantitative Data

Numerical information.

Qualitative Data

Information about qualities or categories.

Discrete Data

Countable data.

Continuous Data

Measurable data.

Frequency

The number of times something occurs.

Bar Chart

Uses bars to compare quantities.

Line Graph

Uses plotted points joined by lines to show changes.

UNIT 14 REVISION SUMMARY

Remember:

- Data is collected information.
 - Quantitative data is numerical.
 - Qualitative data describes qualities or categories.
 - Discrete data is countable.
 - Continuous data is measurable.
 - Tally charts help record frequencies.
 - Bar charts are useful for comparing categories.
 - Line graphs are useful for showing changes over time.
 - Always label graph axes and use a suitable scale.
 - Graphs help us interpret data and make conclusions.
-

END-OF-UNIT REVISION EXERCISE

A. Data Types

1. Define quantitative data.
 2. Give two examples of quantitative data.
 3. Define discrete data.
 4. Define continuous data.
 5. State whether each is discrete or continuous:
 - a) Number of books
 - b) Height of a pupil
 - c) Time taken for a race
 - d) Number of pupils
 - e) Mass of rice
-

B. Tally Charts

6. Twenty pupils choose their favourite fruit:
 - Apple = 6
 - Banana = 5
 - Orange = 5
 - Mango = 4

Draw a tally chart.

7. Which fruit is most preferred?
 8. Which fruit is least preferred?
-

C. Bar Charts

9. A school records pupils in four classes:
 - P2 = 30
 - P3 = 35
 - P4 = 40
 - P5 = 35

Draw a bar chart.

10. Which class has the highest number of pupils?
 11. Which two classes have the same number?
-

D. Line Graphs

12. A plant grows as follows:

Week	Height
1	5 cm
2	8 cm
3	11 cm
4	14 cm

Draw a line graph.

13. Describe the pattern of growth.
 14. If the pattern continues, predict the height in Week 5.
-

E. Interpretation

15. What does a rising line on a line graph show?
16. What does a horizontal line show?
17. What does a falling line show?
18. Give one situation where a bar chart is more useful than a line graph.
19. Give one situation where a line graph is more useful than a bar chart.
20. Explain why collecting and presenting data is useful in daily life.

UNIT 15: INTRODUCTION TO LIKELIHOOD OF EVENTS

Key Unit Competence

By the end of this unit, you should be able to:

- explain the likelihood of events,
 - conduct simple experiments,
 - record outcomes,
 - use probability vocabulary correctly,
 - compare events as more likely, less likely, certain or impossible,
 - appreciate that random events cannot always be predicted.
-

15.0 Introduction

Likelihood describes how possible it is for an event to happen.

Some events:

- will definitely happen,
- may happen,
- may be more likely than others,
- may be less likely,
- may be impossible.

Probability can be explored using:

- tossing coins,

- rolling dice,
 - drawing coloured objects,
 - weather predictions,
 - games of chance.
-

15.1 Outcomes of a Die Toss

A standard die has six faces:

1, 2, 3, 4, 5 and 6

When a fair die is rolled once, any one of these six numbers may appear.

The possible outcomes are:

1, 2, 3, 4, 5, 6

Each face has an equal chance of:

$\frac{1}{6}$.

Important Ideas

Before rolling the die:

The result is uncertain.

After the die is rolled:

The outcome becomes known.

It is impossible for two different faces to be on top at the same time in one roll.

Example

If you predict that a 6 will appear:

Chance of getting 6:

$\frac{1}{6}$

Chance of not getting 6:

$\frac{5}{6}$

Experimental Results

When a die is rolled many times, each result can be recorded using:

- a list,
 - tally marks,
 - a frequency table,
 - a bar chart.
-

Example

A die is rolled 30 times.

Results:

Face	Frequency
1	7
2	5
3	1
4	3
5	6
6	8

The most frequent outcome is:

6

The least frequent outcome is:

3.

Revision Exercise 1

1. List all possible outcomes when a die is rolled.
 2. What is the chance of rolling a 4 on a fair die?
 3. Is it possible to get 2 and 5 at the same time in one roll?
 4. A die is rolled 20 times. Record the results in a tally table.
 5. Represent your results using a bar chart.
-

15.2 Vocabulary of Chance

Important probability words include:

- experiment,
- outcome,
- equal chance,
- certain,
- impossible,
- equally likely,
- less likely,
- likely,
- more likely.

Experiment

An experiment is an activity carried out to find what happens.

Example:

Tossing a coin

Outcome

An outcome is a possible result of an experiment.

For a coin toss, the possible outcomes are:

Head or Tail

Equal Chance

Events have an equal chance when they are equally likely to happen.

Example:

A fair coin has:

Head = equal chance

Tail = equal chance

Certain Event

A certain event will definitely happen.

Example:

The sun will rise.

Impossible Event

An impossible event cannot happen.

Example:

Getting both head and tail at the same time when tossing one ordinary coin.

Equally Likely

Events are equally likely when they have the same chance.

Example:

For a fair die, every face from 1 to 6 is equally likely.

Less Likely

An event is less likely if it has a smaller chance than another event.

Example:

If a bag contains many black cards and very few red cards, picking red is less likely.

Likely

An event is likely when it has a good chance of happening.

Example:

Rain may be likely when the sky is very cloudy.

More Likely

An event is more likely when it has a higher chance than another event.

Revision Exercise 2

Choose the correct vocabulary:

1. The sun rising tomorrow is _____.
 2. Getting 7 on a standard six-sided die is _____.
 3. Getting head or tail on a fair coin are _____.
 4. If a bag contains 8 blue balls and 2 red balls, blue is _____ to be chosen.
 5. Red is _____ to be chosen than blue.
-

15.3 Certain, Uncertain and Impossible Events

Certain Event

A certain event always happens.

Chance:

100%

Uncertain Event

An uncertain event may or may not happen.

Examples:

- it may rain tomorrow,
 - a certain team may win a football match,
 - you may roll a 3 on a die.
-

Impossible Event

An impossible event never happens.

Chance:

0%.

Examples

Event 1

Drop a ball from your hand.

It will fall to the ground.

This is:

Certain

Event 2

Roll a die and get 3.

This is:

Uncertain

Event 3

A cow flies like a bird.

This is:

Impossible.

Revision Exercise 3

Write Certain, Uncertain or Impossible:

1. The sun will rise tomorrow.
 2. It will rain next week.
 3. A stone will grow into a tree.
 4. A pupil will get number 5 when rolling a die.
 5. A dropped book will move towards the ground.
 6. A standard die will show number 10.
-

15.4 Experiments with Coins

A fair coin has two possible outcomes:

Head

And:

Tail

Each outcome has an equal chance.

Therefore:

Head and Tail are equally likely.

Example

Toss a coin 20 times.

Record every outcome.

Throw	Result
1	Head
2	Tail
3	Head
...	...

After all throws:

- count the number of heads,
- count the number of tails,
- compare the results.

The number of heads may not be exactly equal to the number of tails in a small experiment.

But both outcomes still have equal theoretical chances for a fair coin.

Important Point

The outcome of one coin toss does not determine the next outcome.

For example, if the first toss gives Head, the next toss can still be Head or Tail.

Likely and Unlikely

The source uses:

Likely for an event with a good chance of happening.

Unlikely for an event with a low chance of happening.

More likely compares events where one has a higher chance.

Less likely compares events where one has a lower chance.

Example 1

A fair coin is tossed.

Which is more likely: Head or Tail?

Neither.

They are:

Equally likely.

Example 2

A bag contains:

2 red balls

And:

8 blue balls

There are more blue balls.

Therefore:

Blue is more likely to be picked.

Red is:

Less likely to be picked.

Example 3

A bag contains:

- 4 red marbles,
- 1 blue marble.

Red is more likely because there are more red marbles.

Example 4

A die has faces:

1, 2, 3, 4, 5, 6

Odd numbers are:

1, 3, 5

There are three odd-number outcomes.

There is only one outcome for rolling 6.

Therefore:

Rolling an odd number is more likely than rolling a 6.

Revision Exercise 4

1. A bag contains 4 red marbles and 1 blue marble. Which colour is more likely?
 2. There are 12 rainy days and 3 sunny days. Which type of day is more likely?
 3. In a class, 27 pupils prefer pizza and 3 prefer sweets. Which choice is more likely?
 4. A die is rolled. Which is more likely:
 - rolling a 6,
 - rolling an odd number?
 5. A bag has 9 green balls and 1 yellow ball. Which colour is less likely?
-

Probability and Frequency

When an experiment is repeated, we can count how often each result occurs.

This is called its frequency.

For example:

If a die is rolled 48 times:

Face	Frequency
1	9
2	8
3	8
4	7
5	8
6	8

The face with the highest frequency is:

1

The face with the lowest frequency is:

4.

Fair and Random Events

A random event is an event where the exact outcome cannot be known before the experiment.

Examples:

- tossing a coin,
- rolling a die.

A fair game gives outcomes an appropriate equal chance when the game is designed equally.

The source stresses that random events cannot be predicted with certainty.

IMPORTANT LIKELIHOOD RULES

Certain

Will happen.

100% chance

Impossible

Cannot happen.

0% chance

Uncertain

May or may not happen.

Equally Likely

Two or more events have the same chance.

More Likely

One event has a greater chance than another.

Less Likely

One event has a smaller chance than another.

IMPORTANT EXPERIMENT RULES

Fair Coin

Possible outcomes:

Head, Tail

They are equally likely.

Fair Die

Possible outcomes:

1, 2, 3, 4, 5, 6

Each face has equal chance:

$\frac{1}{6}$

UNIT 15 REVISION SUMMARY

Remember:

- Likelihood tells us how possible an event is.
- An experiment is an activity carried out to observe an outcome.
- An outcome is a possible result.
- A fair coin has two equally likely outcomes.
- A fair die has six equally likely outcomes.
- A certain event definitely happens.
- An impossible event cannot happen.
- An uncertain event may or may not happen.
- More likely means greater chance.
- Less likely means smaller chance.

- Repeated experiments can be recorded using tally tables, frequency tables and bar charts.
 - Experimental frequencies may vary from one experiment to another.
 - Random events cannot always be predicted.
-

END-OF-UNIT REVISION EXERCISE

A. Vocabulary

1. Define an experiment.
 2. Define an outcome.
 3. What is a certain event?
 4. What is an impossible event?
 5. What does equally likely mean?
 6. Explain the difference between more likely and less likely.
-

B. Dice

6. List all possible outcomes when a standard die is rolled.

7. What is the chance of rolling a 2?

8. Is rolling a 7 possible on a standard six-sided die?

9. Which is more likely:

- rolling 6,

- rolling an even number?

Explain.

C. Coins

10. What are the two outcomes when a coin is tossed?

11. Are Head and Tail equally likely on a fair coin?

12. Can Head and Tail both face up at the same time on one ordinary coin toss?

13. A coin is tossed 15 times and lands:

- Head = 7 times

- Tail = 8 times

Record the frequencies.

D. Certain, Uncertain or Impossible

14. The sun rises tomorrow.

15. It rains next Tuesday.

16. A cow flies like a bird.

17. A die shows 4 when rolled.

18. A standard die shows 9.

19. A dropped object moves towards the ground.

E. Comparing Likelihood

20. A bag contains:

- 9 red marbles,
- 2 blue marbles,
- 4 green marbles.

Which colour is most likely to be selected?

21. Which colour is least likely?

22. A bag contains 2 yellow balls and 10 black balls. Is yellow more likely or less likely?

23. In a class of 30 pupils, 27 prefer football and 3 prefer basketball. Which choice is more likely?

F. Frequency and Graphs

24. A die is rolled 20 times:

Face| Frequency

1| 3

2| 5

3| 2

4| 4

5| 3

6| 3

- a) Which face appeared most frequently?
 - b) Which face appeared least frequently?
 - c) How many times did face 3 appear?
 - d) Draw a bar graph of the data.
-

G. Real-Life Thinking

- 25. Explain why the result of a random experiment cannot always be predicted.
- 26. Give one real-life example of a certain event.
- 27. Give one real-life example of an uncertain event.
- 28. Give one real-life example of an impossible event.

29. Explain why learning about likelihood is useful in daily life.